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Master of Research in Business Analytics

The Oil Price of Diplomacy

*Causal Evidence from Bilateral Diplomatic Turning Points,
Machine Learning, and Cross-Dyad Network Spillovers*

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DECLARATION

I affirm that this report is my own original work, and any assistance or collaboration has been acknowledged. All sources of information, ideas, or data have been appropriately cited, whether they are directly quoted or paraphrased. Additionally, this report complies with the academic integrity policies and guidelines of TBS.

No part of this report has been submitted for any other degree or qualification at this or any other institution.

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Finally, I am acutely aware that the opportunity to pursue this education is not one that everyone enjoys. I think of those who have been denied access to education by poverty, conflict, disability, or discrimination. The privilege of being able to learn, to conduct research, and to be guided throughout this journey is not lost on me. I hope this work reflects, in some small measure, my appreciation for that privilege.

ABSTRACT

This thesis replicates and extends Saadaoui (2026), who identifies bilateral geopolitical turning points using the second difference of a Political Relationship Index (Δ^2 PRI) and estimates their causal effect on oil prices through instrumental variable local projections. The analysis extends the original China–United States study across three dimensions: measurement, estimation, and external validity.

The **measurement extension** constructs a Composite Geopolitical Index (CGI) from five NLP pipelines. The composite second difference yields a first-stage $F = 7.15$, far below the Δ^2 PRI benchmark of $F = 193.3$, confirming that automated NLP aggregates cannot replace narrative-coded turning-point instruments for causal identification.

The **estimation extension** applies Double/Debiased Machine Learning and Chernozhukov–Hansen IV quantile local projections to the China–United States dyad. The baseline LP-IV estimate at $h = 6$ is $\hat{\beta}_6 = -0.171$; XGBoost DML corroborates the direction ($\hat{\beta}_6 = -0.114$), while Ridge collapses due to over-partialling. CH-IVQR confirms negative quantile effects across the oil-price distribution, establishing that the mean estimate is not tail-driven.

The **external validity extension** estimates panel LP-IV across eleven dyads. Oil-importing dyads produce negative short-run coefficients while the China–Russia dyad is positive ($\beta_6 = +1.102$), consistent with an energy-exporter channel. The Diebold–Yilmaz Total Connectedness Index is 12.53%, indicating that bilateral turning-point shocks are largely dyad-specific rather than driven by a common factor. A causal forest reveals that treatment effects are negatively correlated with diplomatic attention share ($r = -0.803$, $p < 0.001$), confirming a market saturation mechanism.

Keywords: bilateral geopolitical risk, oil prices, instrumental variables, local projections, double machine learning, quantile treatment effects, causal forest, market saturation

JEL Classification: C26, C33, C45, Q43, F51

Executive Summary

This thesis answers the following question: do abrupt improvements in China–United States bilateral relations cause a reduction in global oil prices, and if so, through what mechanism, for how long, and for which bilateral relationships more generally?

The main finding is that a cooperation turning point in the China–United States bilateral diplomatic trajectory reduces the log WTI oil price by approximately 0.17 log points within six months ($\hat{\beta}_6 = -0.171$). This estimate is identified by the second difference of a pre-smoothed Political Relationship Index (Δ^2 PRI), which achieves a first-stage $F = 193.3$ with the extended control set and $F = 236.2$ with the parsimonious Saadaoui baseline. The result is supported by wild bootstrap confidence intervals (which exclude zero at $h = 3$ and $h = 6$) and by CH-IVQR (which confirms negative effects at lower, median, and upper quantiles of the conditional oil-price distribution). The Benjamini-Hochberg false discovery rate correction at $q = 0.10$ renders $h = 6$ non-significant, leaving the short-run inference to rest on the bootstrap and distributional evidence.

Five contributions are reported: (1) a five-source composite GPR index is constructed and shown to be insufficient as an identifying instrument; (2) non-linear confounding is documented via SHAP analysis of XGBoost nuisance functions, motivating the DML extension; (3) distributional homogeneity of the causal effect is established via CH-IVQR; (4) external validity is assessed for eleven dyads using panel LP-IV and pooled DML; and (5) the causal forest saturation result ($r = -0.80$) provides the most original finding: oil markets price geopolitical risk continuously in high-salience dyads, so marginal turning points have smaller effects when diplomatic attention is already saturated. Sub-sample analysis reveals that the geopolitical elasticity nearly doubled after 2010 ($\beta_{pre} = -0.22$ to $\beta_{post} = -0.38$), attributable to the US Shale Revolution’s transformation of WTI into a global marginal barrel.

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List of Abbreviations

AR	Anderson-Rubin (weak instrument robust test)	HTE	Heterogeneous Treatment Effects
ATE	Average Treatment Effect	ICEWS	Integrated Crisis Early Warning System
BDI	Baltic Dry Index	IRF	Impulse Response Function
BERT	Bidirectional Encoder Representations from Transformers	IV	Instrumental Variable
BH	Benjamini-Hochberg (false discovery rate correction)	IVQR	Instrumental Variable Quantile Regression
CAMEO	Conflict and Mediation Event Observations	IVW	Inverse-Variance Weighted
CATE	Conditional Average Treatment Effect	LBMA	London Bullion Market Association
CGI	Composite Geopolitical Index	LP	Local Projection
CH-IVQR	Chernozhukov-Hansen IV Quantile Regression	LP-IV	Local Projection Instrumental Variable
CNY	Chinese Renminbi	MFN	Most-Favoured Nation
DML	Double/Debiased Machine Learning	NLP	Natural Language Processing
DY	Diebold-Yilmaz (connectedness)	NYT	New York Times
EPU	Economic Policy Uncertainty	PLIV	Partially Linear Instrumental Variables
FDR	False Discovery Rate	PRI	Political Relationship Index
FEVD	Forecast Error Variance Decomposition	REER	Real Effective Exchange Rate
FRED	Federal Reserve Economic Data	SHAP	SHapley Additive exPlanations
GAT	Graph Attention Network	SWB	Summary of World Broadcasts (Phoenix)
GDELT	Global Database of Events, Language, and Tone	TCI	Total Connectedness Index
GPR	Geopolitical Risk	TED	Treasury-EuroDollar spread
GSCPI	Global Supply Chain Pressure Index	VAR	Vector Autoregression
HAC	Heteroskedasticity- and Autocorrelation-Consistent	VIF	Variance Inflation Factor
		VIX	Volatility Index (CBOE)
		WITS	World Integrated Trade Solution
		WTI	West Texas Intermediate (crude oil price)
		WTO	World Trade Organization

INTRODUCTION

1.1 Background and Motivation

Geopolitical Risk and Commodity Markets

The relationship between geopolitical conditions and commodity markets has moved to the centre of empirical macroeconomics over the past decade. Oil is particularly sensitive to geopolitical disturbances because its supply is geographically concentrated, its demand is inelastic in the short run, and its futures markets are forward-looking. The precautionary channel, formalised by Kilian (2009), implies that diplomatic events can move oil prices even when no barrel of crude is physically affected. The US-China trade war, the COVID-19 pandemic, and the re-emergence of great-power competition have all coincided with extreme oil price volatility, motivating the need to decompose how much of this volatility is attributable to bilateral diplomatic shocks.

China-Centred Geopolitical Tensions

China occupies a unique position in the global energy system. As the world's largest oil importer, shifts in China's bilateral relations affect global crude oil prices through precautionary stockbuilding, anticipated sanctions, and changes in maritime security. The China–United States relationship is particularly significant: it accounts for the largest share of turning-point shocks in Saadaoui's Political Relationship Index and has undergone the most dramatic deterioration in the sample period. Understanding whether abrupt diplomatic reversals cause oil price movements, not merely correlate with them, requires a credible identification strategy.

Two mechanisms connect bilateral diplomatic turning points to oil prices. The precautionary demand channel implies that cooperation compresses the risk premium, reducing prices. The trade and demand channel implies that improved relations stimulate trade and industrial activity, producing a medium-run positive response. This thesis replicates the baseline strategy of Saadaoui (2026) and extends it across three dimensions: measurement, estimation, and external validity.

1.2 Research Problem and Research Gap

The primary challenge in identifying the causal effect of geopolitical risk on oil prices is endogeneity. Oil price shocks feed back into diplomatic relations, and aggregate risk measures absorb global economic conditions that simultaneously drive commodity markets. The dominant geopolitical risk index, constructed by Caldara and Iacoviello (2022), is aggregate and cannot serve as an instrumental variable.

Even with a valid instrument, the linear LP-IV framework imposes two restrictions. It assumes linear confounding and reports only the average treatment effect, masking heterogeneity across market conditions, distributional segments, and bilateral contexts. A single dyad estimate provides no evidence of external validity. These gaps motivate the machine learning extensions and panel analysis in this thesis.

1.3 Research Questions

This thesis addresses five research questions:

- RQ1.** Can a composite NLP index replicate Δ^2 PRI as an instrument?
- RQ2.** Are LP-IV impulse responses robust to non-linear confounding under DML?
- RQ3.** Does the average effect mask distributional heterogeneity across the WTI distribution?
- RQ4.** Do the findings generalise to other dyads, and does the sign align with each partner's energy role?
- RQ5.** Does the treatment effect vary systematically with diplomatic attention share?

1.4 Research Contributions

Bilateral Geopolitical Measurement: The thesis replicates Saadaoui (2026) in Python, achieving exact parity on first-stage F-statistics ($F = 236.19$ parsimonious, $F = 193.26$ extended) and matching within 0.082 on impulse responses.

Composite Geopolitical Index: A CGI is constructed from five NLP sources (GDELT, ICEWS, Phoenix SWB, Phoenix NYT, FinBERT). The composite Δ^2 CGI yields $F = 7.15$, below the Stock-Yogo (2005) threshold of 16.38 and far below the Δ^2 PRI benchmark of $F = 193.3$, confirming that automated NLP aggregates cannot fully replace narrative instruments for causal identification. Full diagnostics are in Appendix A.

Machine Learning Extensions: DML with XGBoost corroborates the baseline ($\hat{\beta}_6 = -0.114$ vs -0.171), while Ridge collapses due to over-partialling. CH-IVQR confirms negative effects across

all quantiles ($\tau = 0.10$: -0.223 ; $\tau = 0.50$: -0.064 ; $\tau = 0.90$: -0.150), confirming the effect is not tail-driven. Full SHAP decompositions are in Appendix C.

Cross-Dyad Validation: Panel LP-IV across eleven dyads shows oil-importing partners produce negative short-run coefficients, while China–Russia yields a positive coefficient ($\hat{\beta}_6 = +1.102$). The Diebold–Yilmaz Total Connectedness Index is 12.53%, indicating that bilateral turning-point shocks are largely dyad-specific rather than driven by a common factor. Panel results are in Appendix D.

Market Saturation: A causal forest reveals that treatment effects are negatively correlated with ICEWS diplomatic attention share ($r = -0.803$, $p < 0.001$). The geopolitical elasticity nearly doubled after 2010 ($\beta_{pre} = -0.22$ to $\beta_{post} = -0.38$), attributable to the US Shale Revolution. Causal forest diagnostics and break tests are in Appendices D and F.

1.5 Thesis Structure

Chapter 2 reviews the literature. Chapter 3 describes the data and variable construction. Chapter 4 lays out the identification strategy and estimation procedures. Chapter 5 reports the empirical findings. Chapter 6 concludes with policy implications and future research directions. Supporting analyses are collected in Appendices A–G.

Literature Review

2.1 Geopolitical Risk and Oil Prices

2.1.1 *Structural Approaches to Oil Market Shocks*

Geopolitical events and oil prices have attracted sustained empirical attention since the 1970s supply disruptions. Kilian (2009) provides the canonical framework, identifying three structurally distinct shocks: crude oil supply disruptions, aggregate demand shocks, and precautionary demand shocks. The latter is the primary geopolitical-risk channel: a credible signal of future supply disruption raises inventory demand as insurance, bidding up the spot price even absent physical disturbance.

Baumeister and Kilian (2016) extend this framework with time-varying structural uncertainty, showing that ignoring model uncertainty biases geopolitical-shock estimates toward zero. Antonakakis et al. (2017) document pronounced nonlinearity via DCC-GARCH dynamic connectedness: the geopolitical risk–oil volatility relationship intensifies during conflict and attenuates during diplomatic stability. This state-dependence, echoed in Ramey (2016)’s survey of shock propagation, motivates methods such as causal forests that recover heterogeneous treatment effects without a binary regime split.

2.1.2 *Aggregate and Regional Geopolitical Risk Indices*

Caldara and Iacoviello (2022) construct the standard benchmark, the GPR index, from newspaper keyword counts covering military conflict, nuclear threats, terrorist attacks, and war rhetoric. Available at country and global level, it predicts investment, industrial production, and commodity prices, and its keyword-counting methodology has become a template for later indices. Despite this influence, GPR has two limitations motivating bilateral approaches: it is aggregate, conflating bilateral shocks (e.g. a US–China deterioration) with generalised global tension; and it is endogenous to global economic conditions, disqualifying it as an instrument without strong additional assumptions.

Bondarenko et al. (2024) address the aggregation problem with local-language GPR indices, showing that shocks identified from local news sources have significantly larger domestic economic effects than those from Anglosphere media. This motivates constructing a bilateral composite index as a nuisance control in the machine learning specifications below.

2.1.3 *Narrative Bilateral Political Relationship Index (PRI)*

Saadaoui (2026) introduces the Political Relationship Index (PRI), a monthly bilateral measure of diplomatic quality built from systematic monitoring of Chinese government statements, official media, and diplomatic communiqués. Unlike keyword-frequency indices, the PRI captures the relationship's underlying trajectory rather than episodic news volume. The identification strategy instruments on the second difference of a five-month trailing mean, $\Delta^2 \text{PRI}$, isolating abrupt, unpredictable turning points: moments when relations accelerate toward cooperation or reverse toward hostility at a rate not predictable from their prior path.

Applied to the China–United States dyad over 1990–2022 ($n = 386$), Saadaoui (2026) finds that a positive PRI turning point reduces log WTI oil prices at the six-month horizon, with the effect reversing sign at medium-run horizons of 24–36 months, and reports a first-stage F-statistic exceeding conventional weak-instrument thresholds.

2.2 *Political Relationship Index and Bilateral Diplomacy*

2.2.1 *Construction of the PRI and Turning-Point Approach*

The PRI fills a distinct niche, combining expert-coded narrative information with systematic monitoring of official communications. Construction involves multi-step aggregation of government statements, diplomatic press releases, and expert assessments on a conflict-cooperation scale, log-transformed to approximate a continuous Gaussian process; the resulting series is smoother and less prone to noise-driven spikes than event-count indices, though more labour-intensive and limited to dyads with sufficient primary source material.

The identification strategy instruments on the second difference of a five-month trailing mean of the log PRI. Let $\tilde{\ell}_t$ denote the five-month trailing mean:

$$\tilde{\ell}_t = \frac{1}{5} \sum_{k=0}^4 \ell_{t-k} \text{Five-month trailing mean of log PRI} \quad (2.1)$$

The second-difference operator isolates abrupt, unpredictable turning points:

$$\Delta^2 \text{PRI}_t \equiv \tilde{\ell}_t - 2\tilde{\ell}_{t-1} + \tilde{\ell}_{t-2} \text{Second difference of the Political Relationship Index} \quad (2.2)$$

This transformation is zero for any linear trend or level change and non-zero only when the

trajectory changes direction unexpectedly. The instrument is thus insensitive to gradual drifts, responding only to sharp, discrete reversals (an ambassador recalled, a sudden trade concession, a joint communiqué ending a territorial dispute) far more plausibly exogenous to oil market fundamentals than gradual trends that co-evolve with trade volumes and energy demand.

2.2.2 *Comparison with Aggregate GPR Measures*

The Δ^2 PRI instrument exhibits substantially stronger first-stage properties than aggregate GPR indices. This gap reflects its turning-point structure: automated sources capture aggregate stress levels but not the acceleration-reversal structure the second-difference transformation requires to isolate variation plausibly orthogonal to economic conditions, the distinction this thesis tests directly when assessing whether automated composites can approximate it.

2.3 Text-Based Measurement of Geopolitical Events

The last decade has seen an explosion of automated, large-scale text-based datasets for tracking political events, providing high-frequency, geographically disaggregated measures that complement traditional narrative approaches. The literature draws on five principal sources for automated geopolitical event measurement.

2.3.1 *GDELT and ICEWS*

The GDELT database (Leetaru and Schrod, 2013) covers geopolitical events from 1979 to the present, automatically coding news articles from thousands of global sources using the CAMEO taxonomy and Goldstein scale (Goldstein, 1992). Its strength is breadth, near-universal coverage and daily granularity from 2013, but it aggregates indiscriminately across sources, quality, and languages, so the resulting series can be dominated by article volume rather than diplomatic significance.

The ICEWS (Boschee et al., 2013) instead combines automated coding with curated source lists and human quality control. Covering 1995 onward, with Goldstein scores validated against military conflict outcomes, its curated nature yields higher precision than GDELT at the cost of reduced temporal coverage and geographic scope.

2.3.2 *Phoenix Event Data and New York Times Archive*

The Phoenix event data system (Schrod and Beiler, 2019) provides two complementary CAMEO-coded subsets: the Summary of World Broadcasts (SWB) panel, processing BBC Monitoring transcripts for a non-US-centric perspective (1990–2019), and the New York Times (NYT) archive covering China–United States from 1945 onward (used here for 1990–2018). Their independence from GDELT

makes them valuable for cross-validation: if GDELT, ICEWS, SWB, and NYT independently flag the same high-stress episodes, the composite signal is validated across methodologically distinct pipelines. Bagozzi and Schrodtt (2021) extend the NYT archive with additional validation, showing NYT coverage of US–China relations correlates strongly with known diplomatic turning points.

2.3.3 *Transformer-Based Sentiment: FinBERT*

Araci et al. (2019) apply BERT-style pre-training to financial text, producing FinBERT, a fine-tuned sentiment classifier distinguishing positive, negative, and neutral tone, outperforming general-purpose models on domain-specific sentiment classification. FinBERT complements event-count measures by capturing the tone of coverage rather than just its volume – an important distinction, since high volume may reflect either intense diplomatic activity or routine reporting, whereas sentiment captures the relationship’s qualitative direction.

2.3.4 *Strengths and Limitations of Automated Sources*

Three structural limitations are shared across automated event databases: automated coding conflates event volume with importance; CAMEO coding is trained primarily on English-language media, biasing toward US-centric framing; and automated measures capture aggregate stress but not the acceleration-reversal structure causal identification requires from second-difference transformations. These limitations motivate using automated sources as nuisance controls rather than primary instruments.

2.4 Causal Inference in Macroeconomics

2.4.1 *Local Projections*

Jordà (2005) introduces local projections as an alternative to VAR-based impulse response estimation. Rather than imposing a single dynamic model, local projections estimate a separate reduced-form regression for each horizon h :

$$y_{t+h} = \alpha_h + \beta_h d_t + \gamma_h(L)\mathbf{x}_t + \varepsilon_{t+h} \text{LocalProjectionequation} \quad (2.3)$$

yielding the horizon- h response β_h directly: mis-specification at one horizon does not contaminate estimates at others, and the approach extends trivially to instrumental variables and panel settings (Montiel Olea and Plagborg-Møller, 2021). Because overlapping forecast errors ε_{t+h} follow an $\text{MA}(h - 1)$ process, valid inference requires HAC standard errors with bandwidth proportional to h (Newey and West, 1987).

2.4.2 *Instrumental Variables and Weak Instruments*

The LP-IV estimator instruments the endogenous d_t with z_t via two-stage least squares at each horizon. Validity requires relevance (diagnosed by the first-stage F-statistic) and exogeneity (the exclusion restriction). Stock and Yogo (2005) provide critical values for the maximum-eigenvalue F-statistic bounding size distortions; Montiel Olea and Pflueger (2013) develop more robust critical values for heteroskedastic, autocorrelated settings. Anderson and Rubin (1949) provide an early weak-instrument inference approach still useful as a diagnostic when the first-stage F falls below conventional thresholds.

2.5 *Machine Learning for Causal Inference*

2.5.1 *Double/Debiased Machine Learning*

Chernozhukov et al. (2018) develop the DML framework to address a key problem: when nuisance functions are estimated with regularised machine learning, regularisation bias propagates into the causal estimate and does not vanish asymptotically. DML eliminates this bias through Neyman orthogonality and cross-fitting. Applied to LP-IV, DML produces residualised series $\tilde{Y}_{t+h} = y_{t+h} - \hat{g}(\mathbf{x}_t)$, $\tilde{D}_t = d_t - \hat{m}(\mathbf{x}_t)$, and $\tilde{Z}_t = z_t - \hat{r}(\mathbf{x}_t)$, recovering the causal parameter from the just-identified IV regression of $\tilde{Y}$ on $\tilde{D}$ using $\tilde{Z}$ – equivalent to relaxing linearity on the control function while preserving valid IV identification.

A critical consideration is the observations-to-variables ratio (n/p), governing overfitting risk and cross-fitting feasibility. With limited effective observations, n/p often approaches the lower bound for reliable nonparametric nuisance estimation, constraining model depth. Gonçalves and Kilian (2004) develop bootstrap methods for autoregressions with conditional heteroskedasticity that improve small-sample inference relative to asymptotic approximations.

2.5.2 *Causal Forests and Heterogeneous Treatment Effects*

Wager and Athey (2018) and Athey et al. (2019) develop causal forests for estimating conditional average treatment effects (CATEs), adapting random forests to directly maximise treatment-effect heterogeneity. Honestly, separating the sample used to grow trees from the sample used to estimate leaf-level effects, ensures consistent and asymptotically normal CATEs under mild regularity conditions.

Ramey and Zubairy (2018) use a split-sample approach to estimate state-dependent fiscal multipliers via LP-IV, finding multipliers vary significantly across regimes. The causal forest framework subsumes this by treating the moderator as continuous rather than imposing a binary split, valuable when the

effective subsample would otherwise fall below the minimum needed for stable IV estimation. Jentsch and Lunsford (2019) challenge the validity of the residual-based wild bootstrap for proxy-SVAR inference, proposing a moving block bootstrap alternative, a direct precedent for the wild bootstrap robustness check applied in Section 4.9.1.

2.6 Distributional Treatment Effects

Chernozhukov and Hansen (2005) develop an IV estimator for structural quantile treatment effects (IVQR) under rank similarity. Their Algorithm 1 identifies $\alpha(\tau)$ by grid-search inversion: for each candidate α , the instrument should have zero predictive power in the conditional quantile regression of the rank-adjusted outcome; the value minimising the absolute t-statistic constitutes the IVQR estimate, with confidence sets obtained by test inversion.

Structural quantile effects complement average effects by revealing whether the average masks heterogeneity across the outcome distribution, and serve as a robustness check: a genuine structural effect produces a systematically signed quantile function, whereas an average driven by a few extreme observations would not. Benjamini and Hochberg (1995) develop the false discovery rate (FDR) approach to multiple testing, relevant when many hypotheses are tested simultaneously.

2.7 Network Analysis of Geopolitical Relations

2.7.1 *Diebold–Yilmaz Connectedness*

Diebold and Yilmaz (2012) develop a connectedness measure based on forecast error variance decompositions (FEVDs) from a VAR model. The total connectedness index summarises the share of forecast error variance attributable to shocks from other series, with a TCI near 100% indicating a system dominated by common shocks rather than idiosyncratic variation. Widely applied to financial and macroeconomic networks, its key advantage is isolating directional spillovers, revealing which units are net transmitters or receivers.

2.7.2 *Graph Attention Networks*

Veličković et al. (2018) introduce Graph Attention Networks (GATs), which learn attention-weighted message-passing between graph nodes, assigning differential importance to neighbours rather than treating them uniformly. A cautionary note: GATs can yield uninformative uniform weights when a dominant common factor overwhelms idiosyncratic node features.

2.8 Research Gap and Positioning of this Study

The preceding review identifies four gaps. **Gap 1: Bilateral NLP composites as instruments.** No prior study has systematically tested whether automated multi-source composites can achieve the first-stage power required for causal IV identification, despite the documented limitations of aggregate GPR indices (Caldara and Iacoviello, 2022). **Gap 2: Non-linear confounding robustness.** Saadaoui (2026)’s baseline assumes linear nuisance functions; DML (Chernozhukov et al., 2018) has not been applied to bilateral diplomatic shocks on oil prices. **Gap 3: Distributional heterogeneity.** IVQR (Chernozhukov and Hansen, 2005) has not been applied to geopolitical oil-price transmission, leaving distributional effects unexplored. **Gap 4: External validity across dyads.** Whether Saadaoui (2026)’s findings generalise to other China-centred dyads remains unknown.

These gaps motivate the empirical strategy developed in Chapter 4. The results in Chapter 5 address each gap in turn.

2.9 Summary of the Literature

Tables 2.1 and 2.2 provide a structured comparison of the principal contributions most closely related to this thesis, organised respectively by methodological/econometric contribution and by empirical substance or data source.

Table 2.1: Methodological and econometric contributions

Study	Method	Data / Sample	Key Finding	Gap Addressed
Jordà (2005)	Local Projections (LP)	Macro time series	Flexible IRF estimation directly at each horizon; accommodates IV	No built-in identification for endogenous shocks; assumes linearity
Chernozhukov and Hansen (2005)	IV Quantile Regression	Theoretical + applications	Identifies structural treatment effects under rank similarity	Computationally intensive; not yet applied to geopolitical oil-price transmission
Chernozhukov et al. (2018)	Double/Debiased ML	High-dimensional settings	Removes regularisation bias via Neyman orthogonality	Requires large n/p ratio; not yet applied to bilateral diplomatic shocks
Wager and Athey (2018)	Causal Forests	Theoretical / micro data	Honest estimation of heterogeneous treatment effects (CATE)	Needs adaptation for time-series dependence and diplomatic-attention moderators
Diebold and Yilmaz (2012)	FEVD Connectedness	Financial time series	Quantifies total and directional spillovers	Assumes stationarity; sensitive to lag selection and common factors

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Study	Method	Data / Sample	Key Finding	Gap Addressed
Veličković et al. (2018)	Graph Attention Networks	Graph-structured data	Learns differential attention weights between nodes	Collapses with homogeneous inputs / dominant common factors
Montiel Olea and Pflueger (2013)	Weak-IV robust inference	Theoretical simulations	+ Robust critical values for weak instruments under HAC errors	Purely diagnostic; no application to geopolitical turning points
Anderson and Rubin (1949)	Weak-IV inference	Theoretical	Early approach to inference with weak instruments	Diagnostic only; requires extension to panel time-series settings
Gonçalves and Kilian (2004)	Bootstrap inference	Time series	Bootstrap methods improve inference in small samples	Methodological; not applied to geopolitical oil-price transmission
Jentsch and Lunsford (2019)	Wild bootstrap critique (proxy-SVAR)	US tax data	Residual-based wild bootstrap can yield invalid (too-narrow) confidence intervals	Motivates the wild-bootstrap validity check in Section 4.9.1; not yet tested in LP-IV settings

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Study	Method	Data / Sample	Key Finding	Gap Addressed
Benjamini and Hochberg (1995)	FDR multiple testing	Theoretical	Controls false discovery rate in multiple testing	Methodological; requires application to impulse response testing
Ramey (2016)	Handbook survey	Macroeconomic shocks	Comprehensive review of shock propagation mechanisms	Theoretical survey; no empirical application to bilateral diplomacy

Table 2.2: *Empirical studies and measurement sources*

Study	Method	Data / Sample	Key Finding	Gap Addressed
Kilian (2009)	Structural VAR	Monthly oil data, 1973–2006	Decomposes oil price shocks into supply, aggregate demand, and precautionary demand	No bilateral identification; precautionary demand conflated with aggregate risk
Caldara and Iacoviello (2022)	Newspaper keyword count (GPR)	Global/country-level, 1900–2020	Aggregate GPR predicts investment, production, and commodity prices	Aggregate measure; endogenous to global conditions; weak as IV
Antonakakis et al. (2017)	DCC-GARCH + TVP	GPR and oil volatility, 1995–2013	Strong time-varying connectedness between GPR and oil prices	Correlation-based; no causal identification; aggregate focus
Baumeister and Kilian (2016)	Time-varying SVAR	Oil market data	Geopolitical effects highly sensitive to model uncertainty	Focus remains aggregate; limited bilateral analysis
Bondarenko et al. (2024)	Local-language news-based GPR	Russia / Anglo-sphere media, sanctions period	Local-language GPR shocks have significantly larger domestic economic effects than Anglosphere-sourced shocks	Single-country case study; bilateral generalisation across dyads untested

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Study	Method	Data / Sample	Key Finding	Gap Addressed
Saadaoui (2026)	LP-IV Δ^2 PRI	with China–USA dyad, 1990–2022	Narrative Δ^2 PRI yields strong instrument; negative short-run effect on oil prices	Single dyad; linear nuisance; no distributional or panel extension
Ramey and Zubairy (2018)	State-dependent LP-IV	US fiscal policy, long sample	Multipliers vary significantly across regimes	Binary regime split; lower power for continuous heterogeneity
Leetaru and Schrod (2013)	GDELT (automated CAMEO)	Global news, 1979–present	Large-scale high-frequency event data with Goldstein scores	High noise; volume-driven; lacks turning-point sensitivity
Boschee et al. (2013)	ICEWS (curated CAMEO)	Global events, 1995–2022	Higher-precision event data; validated against conflict	Coverage starts 1995; weaker first-stage properties than narrative PRI
Schrod and Beiler (2019)	Phoenix Event Data (SWB + NYT)	CAMEO-coded events, 1945–2019	Non-Western perspective on international events	Limited post-2019 coverage; manual validation still needed

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Study	Method	Data / Sample	Key Finding	Gap Addressed
Bagozzi and Phoenix Schrodt (2021)	NYT extension	China–USA, 1945–2018	Extended NYT archive with diplomatic validation	Limited to US–China; no causal identification
Araci et al. (2019)	FinBERT senti- ment analysis	Financial news corpora	High-accuracy domain- specific sentiment classifica- tion	Applied only as input feature; not used for causal identifica- tion
Goldstein (1992)	CAMEO event coding	Theoretical	Establishes conflict- cooperation scoring scale	Foundational scale; no appli- cation to causal inference

The two tables reveal three systematic patterns. First, papers achieving strong causal identification do so through narrative or structural identification, not automated text sources. Second, DML and causal forest methods require strict enforcement of parsimony regarding the n/p ratio before being applied to macroeconomic time series. Third, network methods capture systemic co-movement but can yield uninformative structures when a dominant common factor is present.

Data and Index Construction

3.1 Saadaoui (2026) Replication Dataset

The primary dataset is the publicly available replication package of Saadaoui (2026). It contains monthly observations from January 1990 to February 2022 ($T = 386$) for eleven bilateral country pairs involving China. For each dyad d , the dataset provides the log Political Relationship Index $\ell_{d,t} \equiv \text{lpri}_{d,t}$, constructed from systematic monitoring of Chinese government statements and official media coverage of bilateral diplomatic events.

Table 3.1 reports descriptive statistics for the log PRI series across the eleven dyads.

Table 3.1: *Descriptive statistics of bilateral log PRI series, 1990:01–2022:02*

Dyad	N	Mean	Std. Dev.	Min	Max
CHN-USA	386	0.022	1.268	−2.813	1.909
CHN-JPN	386	0.343	1.337	−1.792	1.792
CHN-AUS	386	1.617	0.417	−0.693	2.041
CHN-FRA	386	1.676	0.470	0.470	2.079
CHN-DEU	386	1.779	0.256	1.253	2.079
CHN-GBR	386	1.458	0.498	−0.916	1.902
CHN-IND	386	1.138	0.831	−1.253	1.932
CHN-IDN	386	1.198	0.708	−1.548	1.825
CHN-PAK	386	2.070	0.055	1.988	2.140
CHN-RUS	386	2.018	0.354	0.262	2.230
CHN-VNM	386	1.295	1.053	−2.054	1.974

Source: Saadaoui (2026) replication data; author’s calculations.

The PRI series is log-transformed by Saadaoui (2026) to approximate a continuous Gaussian process. This transformation stabilises variance and facilitates the construction of a stationary instrument. The raw PRI scores from expert coding are positively skewed, so the log transformation compresses the right tail and makes the series amenable to linear time-series modelling.

3.2 Composite Geopolitical Index (CGI)

To evaluate whether automated text-based measures can approximate the turning-point identification power of the narrative-coded PRI, we construct a Composite Geopolitical Index (CGI) by aggregating five independent measurement pipelines. The rationale for a composite index is that each automated source captures a different facet of geopolitical relations. GDELT provides breadth, ICEWS provides precision, Phoenix provides non-US perspectives, and FinBERT provides sentiment. By combining them, we aim to extract a common signal of bilateral geopolitical tension while averaging out source-specific noise.

Table 3.2 summarises the component sources.

Table 3.2: *Composite geopolitical index: component data sources and key limitations*

Source	Measurement	Coverage	Key limitation
GDELT	Automated tone/volume scoring	All 11 dyads, 1990–2022	High noise; volume-sensitive
ICEWS	Goldstein-scaled event coding	All 11 dyads, 1995–2022	Starts 1995
Phoenix SWB	Structured event coding	All 11 dyads, 1990–2019	Ends 2019
Phoenix NYT	NYT archive coding	CHN-USA only, 1990–2018	Single dyad; ends 2018
FinBERT	Transformer senti- ment	All 11 dyads, 2013–2022	Starts 2013

Source: author’s calculations.

3.2.1 Data Preparation and Harmonisation

Each source is converted to a signed conflict score where positive values indicate hostility and negative values indicate cooperation. The sign convention ensures consistency across sources because higher values always correspond to more adversarial bilateral relations.

For GDELT, we use the net-conflict ratio: $\text{net}_{d,t} = \frac{\text{conflict}_{d,t} - \text{cooperation}_{d,t}}{\text{total}_{d,t}}$. This normalises for the volume of reporting, which prevents months with high news volume from dominating the index.

For ICEWS, Phoenix SWB, and Phoenix NYT, we use the Goldstein score, which is a pre-existing conflict-cooperation scale ranging from -10 (most cooperative) to $+10$ (most hostile) (Goldstein, 1992). We negate this score so that positive values represent hostility, consistent with the GDELT convention.

For FinBERT, we use the non-neutral sentiment mean from the ‘hakonmh/sentiment-xdistil-uncased’ model. The sentiment score ranges from -1 (negative and conflictual) to $+1$ (positive and cooperative). We negate this score so that positive values indicate conflictual sentiment.

3.2.2 *Standardisation and Aggregation*

Raw scores from different sources exhibit substantially different scales. For example, FinBERT scores are bounded between -1 and $+1$, while GDELT net-conflict ratios are bounded between -1 and $+1$ but with different variance properties. ICEWS Goldstein scores range from -10 to $+10$. Directly averaging these raw scores would give disproportionate weight to sources with larger numerical ranges.

To address this, we apply within-dyad z-score standardisation to each source:

$$s_{d,t}^{\text{std}} = \frac{s_{d,t} - \bar{s}_d}{\sigma_{s_d}} \quad (3.1)$$

where $\bar{s}_d$ and σ_{s_d} are the dyad-specific mean and standard deviation. This transformation places each source on a common mean-zero and unit-variance scale, which ensures that no source dominates the composite due to its numerical range.

Winsorisation at ± 3 standard deviations is applied to mitigate extreme artefacts in low-volume months. Winsorisation caps extreme values rather than discarding them, which preserves the information content of the observation while preventing outliers from unduly influencing the composite.

The final CGI is the unweighted mean of the available standardised source scores:

$$\text{CGI}_{d,t} = \frac{1}{N_{d,t}} \sum_{i=1}^5 w_{i,d,t} \cdot s_{i,d,t}^{\text{std}} \quad (3.2)$$

where $w_{i,d,t}$ is an indicator for whether source i is available for dyad d in month t , and $N_{d,t}$ is the number of available sources. The unweighted mean is chosen because there is no a priori reason to privilege one source over another. The composite is designed to extract the common signal across independent measurement pipelines.

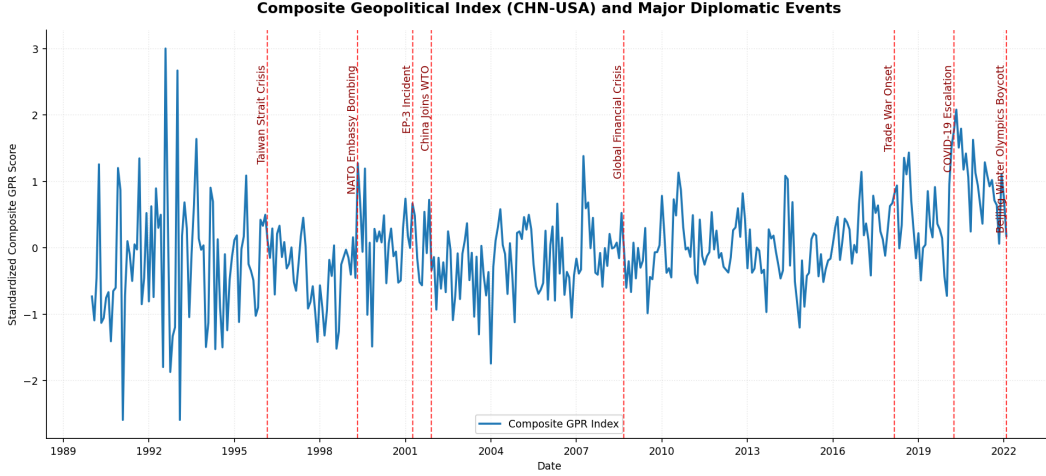


Figure 3.1: Composite geopolitical index (CHN-USA) constructed from five independent sources. Vertical dashed lines mark major diplomatic events. The index registers periods of elevated stress but lacks the sharp turning-point sensitivity of the narrative-coded PRI. This motivates the continued use of Δ^2 PRI as the primary instrument.

3.2.3 CGI Validation: CHN-USA

Figure 3.1 plots the CGI against major bilateral events in the US-China relationship. The index captures broad trends and rises during periods of tension such as the 1999 Belgrade embassy bombing, the 2001 EP-3 spy plane incident, and the 2018 trade war escalation. However, the index is relatively smooth and does not exhibit the sharp reversals characteristic of the Δ^2 PRI instrument. This smoothness reflects the averaging across sources because sudden diplomatic turning points are diluted by the presence of other sources that may not register the same event. This observation is consistent with the weaker first-stage properties of the CGI documented below.

3.3 Instrument Construction

The identification strategy follows Saadaoui (2026) exactly. The goal is to construct an instrument that isolates exogenous variation in diplomatic relations, specifically sharp and unpredictable turning points in the diplomatic trajectory. The instrument is constructed from the log Political Relationship Index $\ell_{d,t}$ in two steps.

3.3.1 Step 1: Pre-smoothing

The first step applies a five-month trailing mean to the log PRI:

$$\tilde{\ell}_{d,t} = \frac{1}{5} \sum_{k=0}^4 \ell_{d,t-k} \quad (3.3)$$

Why a five-month trailing mean? The raw PRI series contains high-frequency noise, which includes

month-to-month fluctuations that reflect reporting irregularities, minor diplomatic incidents, or measurement error. The five-month trailing mean smooths out these transitory fluctuations and captures the underlying diplomatic trajectory. The choice of five months follows Saadaoui (2026), who selected this window to balance noise reduction against the preservation of genuine turning points. A shorter window such as three months would retain more noise, whereas a longer window such as twelve months would attenuate genuine turning points. Five months is a conventional choice in applied macroeconomics for capturing medium-term trends in monthly data.

3.3.2 Step 2: Second Difference

The second step takes the discrete second derivative of the smoothed series:

$$\Delta^2 \text{PRI}_{d,t} = \tilde{\ell}_{d,t} - 2\tilde{\ell}_{d,t-1} + \tilde{\ell}_{d,t-2} \quad (3.4)$$

What does the second difference measure? The second difference measures the acceleration of the diplomatic relationship, which is the rate at which the trajectory is changing direction. To understand this, consider three cases:

1. **Linear trend:** If the relationship is steadily improving or deteriorating at a constant rate, the second difference is zero. For example, if $\tilde{\ell}_t = 0.01t$, then $\Delta^2 \tilde{\ell}_t = 0$. The instrument is insensitive to gradual diplomatic drifts.
2. **Level shift:** If the relationship suddenly jumps to a new level, such as a one-off diplomatic event, the second difference returns to zero after two periods. The instrument captures the abruptness of the change but does not respond to permanent level shifts.
3. **Turning point:** If the relationship changes direction, for example from improving to deteriorating, the second difference is non-zero. This is the key insight because the instrument is only active when the trajectory of the relationship changes direction in an unexpected manner.

Formally, the second difference is zero under any linear trend of the form $\tilde{\ell}_t = a + bt$. It is also zero for any level shift after two periods. It is non-zero only when the trajectory changes curvature. This means the instrument is insensitive to gradual diplomatic drifts or sustained deterioration and responds only to sharp and discrete reversals.

Why does this matter for identification? Gradual diplomatic trends are likely endogenous to oil markets because improving relations may coincide with increasing trade volumes, which affect oil

demand. Sharp and discrete reversals, such as an ambassador recalled following a hacking accusation, a sudden trade concession after months of escalation, or a joint communiqué ending a territorial dispute, are far more plausibly exogenous to oil market fundamentals. The second difference isolates these plausibly exogenous events.

The same procedure is applied to the CGI to produce $\Delta^2\text{CGI}_{d,t}$ for comparison. The composite CGI is retained only as a nuisance control (`cgi_11`) in the DML specification because its first-stage strength is substantially weaker than that of the narrative $\Delta^2\text{PRI}$.

3.4 Macroeconomic and Financial Controls

A rich set of macroeconomic and financial controls is merged with the core dataset to satisfy the ignorability condition in the machine learning stages. The purpose of these controls is to account for confounding factors that may simultaneously affect both diplomatic relations and oil prices. For example, a global economic downturn could reduce oil demand and lower prices while also affecting diplomatic relations through trade tensions. Without controlling for such factors, the estimated effect of diplomatic shocks on oil prices would be biased.

All controls are lagged by one month to prevent look-ahead bias. At time t , the model only uses information available at $t - 1$:

$$\mathbf{x}_t = \{x_{t-1}^{(1)}, x_{t-1}^{(2)}, \dots, x_{t-1}^{(K)}\} \quad (3.5)$$

This ensures that the controls are predetermined with respect to the treatment and outcome, which is a standard requirement for causal identification in time-series settings.

Selected volatility measures, notably `cny_vol`, are winsorised at the 99th percentile to mitigate the impact of extreme revaluation events. Winsorisation caps extreme values at the 99th percentile rather than discarding them. This prevents a small number of extreme observations from driving the results while preserving the information content of the tail observations.

Table 3.3: *Macroeconomic and financial controls: sources and construction details*

Variable	Source	Density	Construction
vix, dvix	Yahoo Finance	Dense	Monthly mean; first difference
gs10, tb3ms	FRED	Dense	10Y Treasury; 3-month T-bill
term_spread	FRED	Dense	GS10 – TB3MS
tedrate	FRED	Dense	TED spread
lreer	FRED (RBUSBIS)	Dense	Log real effective exchange rate
dcny_usd	FRED (DEXCHUS)	Dense	First diff. of log CNY/USD
cny_vol	FRED (DEXCHUS)	Dense	3-month rolling std of dcny_usd (winsorised at 99th percentile)
lgold	LBMA	Dense	Log monthly gold price
brent_wti	FRED	Dense	log(Brent) – log(WTI), lagged
lbdi	Investing.com	Dense	Log Baltic Dry Index
lip	FRED (INDPRO)	Dense	Log US industrial production
tariff_chn	WITS	Dense	China MFN tariff rate (interpolated)
trade_war	Constructed	Dense	Binary: 1 from 2018-03 onward
lepu	policyuncertainty.com	Dense	Log global Economic Policy Uncertainty
gscpi	NY Fed	Sparse	Global Supply Chain Pressure Index

Dense: $\geq 80\%$ coverage. Sparse: robustness specifications only.

Note: tariff_chn is excluded from main X_t due to high VIF; trade_war enters the CHN-USA specification.

Why these controls? The controls are chosen to capture the main channels through which omitted variables could bias the estimated effect of diplomatic shocks on oil prices.

The **financial controls**, which include VIX, term spread, and TED spread, capture global financial conditions that affect both the cost of holding oil inventories and the risk appetite of commodity traders. The **exchange rate controls**, which include REER, CNY/USD, and CNY volatility, capture the role of the US dollar in oil pricing because oil is priced in dollars and dollar strength affects oil prices independently of diplomatic relations. The **commodity controls**, which include gold, Brent-WTI spread, and BDI, capture global commodity market conditions and shipping costs that affect oil prices through supply chains. The **real activity controls**, which include industrial production, EPU, and GSCPI, capture

global economic conditions that affect oil demand. The **trade policy controls**, which include `tariff_chn` and `trade_war`, capture explicit trade policy changes that may be correlated with both diplomatic relations and oil prices.

Trade policy controls. China's simple-average MFN applied tariff rate from WITS is included as a time-varying control. The series declines sharply after WTO accession in December 2001. A binary `trade_war` indicator, which equals one from March 2018 onward, is added for the US-China escalation period. The tariff series is excluded from the main X_t matrix due to high variance inflation factor (VIF), which indicates collinearity with other controls. It is retained for robustness checks.

Data density. Controls are classified as *dense* if they have $\geq 80\%$ coverage across the sample. Dense controls enter the main specification. Sparse controls, which include EPU and GSCPI, are retained only for robustness specifications where the reduction in sample size is acceptable.

This completes the data construction. The resulting master panel forms the basis for all subsequent empirical analyses in Chapters 4 and 5.

Methodology

4.1 Empirical Strategy Overview

The empirical objective of this thesis is to establish a causal link between bilateral geopolitical turning points and oil price dynamics while addressing key limitations in the existing literature. The research workflow proceeds in five sequential stages:

1. **Baseline Replication:** Replicating the linear LP-IV framework of Saadaoui (2026) to establish a benchmark and validate the Δ^2 PRI instrument.
2. **Measurement Extension:** Constructing a Composite Geopolitical Index (CGI) from five independent NLP sources to test whether automated measures can serve as instruments or nuisance controls.
3. **Non-linear Robustness:** Applying Double/Debiased Machine Learning (DML) to flexibly partial out high-dimensional macroeconomic confounders and relax linearity assumptions.
4. **Distributional Analysis:** Examining whether average effects mask heterogeneity across the oil price distribution using Chernozhukov and Hansen (2005) IVQR.
5. **Heterogeneity and Spillovers:** Exploring conditional treatment effects with Causal Forests and systemic connectedness with the Diebold and Yilmaz (2012) network analysis framework.

This layered design allows us to assess the sensitivity of results to functional form assumptions, control selection, and effect heterogeneity across the oil price distribution and across dyads. Each layer builds upon the previous one, providing a comprehensive robustness check of the baseline findings.

4.2 Identification Strategy and Assumptions

The core identification challenge is the endogeneity of bilateral relations: diplomatic trajectories and oil demand often co-evolve with the global business cycle. We follow the turning-point identification strategy of Saadaoui (2026), using the second difference of the Political Relationship Index (Δ^2 PRI) as an instrument. The construction of this instrument is detailed in Section 3.3.

4.2.1 *Baseline LP-IV Assumptions*

Identification in the LP-IV framework rests on three primary assumptions:

- **Instrument Relevance:** The second difference of the smoothed PRI ($\Delta^2\text{PRI}$) must strongly correlate with the level of the PRI. This is verified by first-stage F-statistics. The strong first-stage is a direct consequence of the turning-point structure of the instrument, which captures abrupt changes in diplomatic trajectory that are not predictable from prior trends. Following Stock and Yogo (2005), we require $F > 10$ for weak-instrument diagnostics; following Montiel Olea and Pflueger (2013), we also report robust critical values for heteroskedastic and autocorrelated settings.
- **Exclusion Restriction:** The abrupt reversals captured by $\Delta^2\text{PRI}$ must affect WTI prices only through the diplomatic level, not through other macro channels. Because the instrument isolates unpredictable changes in acceleration, it is plausibly uncorrelated with lagged oil market fundamentals. We test this indirectly by examining whether the instrument predicts oil prices before the turning point occurs (lead tests) and by checking whether the results are robust to the inclusion of additional controls.
- **Exogeneity:** Turning points in diplomacy are assumed to be shocks to the acceleration of stress, making them plausibly unpredictable and uncorrelated with lagged oil market fundamentals. This assumption is the most critical and is tested indirectly through robustness checks in Chapter 5, including the inclusion of the CGI as a nuisance control and the application of DML to relax linearity assumptions.

4.2.2 *Assumptions Underlying Double Machine Learning*

The transition to DML introduced in Section 4.5 requires the following assumptions:

- **Neyman Orthogonality:** The moment conditions are designed to be locally insensitive to errors in estimating the nuisance functions. This property, formalised by Chernozhukov et al. (2018), allows consistent estimation of the causal parameter even when the nuisance functions are estimated with machine learning methods that introduce regularization bias. The orthogonality condition ensures that the first-order effects of nuisance estimation error vanish asymptotically.
- **Sample Splitting and Cross-Fitting:** To prevent overfitting of nuisance parameters, the sample is divided using time-series cross-validation. We implement five-fold cross-fitting, where the

data is split into five temporal blocks. For each fold, nuisance functions are estimated on four blocks and the causal parameter is estimated on the held-out block. This ensures that the nuisance functions are estimated out-of-sample relative to the data used for causal estimation, eliminating the overfitting bias that would otherwise arise from using the same observations to estimate both nuisances and the causal parameter.

- **Regularity Conditions:** These include approximate sparsity in the high-dimensional control set, allowing the learners (Ridge and XGBoost) to converge at rates sufficient for valid inference. Specifically, we require that the number of controls p is small relative to the effective sample size n such that $n/p \gg 1$. With $n = 337$ and $p = 15$ after selection, $n/p \approx 22$, which is near the conventional lower bound for reliable nonparametric nuisance estimation.

4.2.3 Identification Assumptions for IV Quantile Regression

The Chernozhukov and Hansen (2005) IVQR framework, applied in Section 4.6, relies on the following identification conditions:

- **Monotonicity:** The endogenous treatment must move monotonically with the instrument across the conditional distribution. This means that a positive shock to $\Delta^2\text{PRI}$ must shift the entire distribution of lpri in the same direction, without reversing the ordering of units. This assumption is standard in IVQR and ensures that the quantile effects are identified.
- **Rank Similarity:** The relative rank of an observation in the outcome distribution is stable under different treatment levels. This assumption allows us to interpret the quantile effects as structural rather than selection effects. In practice, this means that observations at the τ -th quantile of the outcome distribution remain at the τ -th quantile regardless of the treatment level, after conditioning on observables.

4.2.4 Assumptions for Causal Forest Estimation

The Causal Forest DML implementation, detailed in Section 4.7, requires the following conditions:

- **Conditional Independence:** Given the moderators and controls, the treatment is as-good-as-randomly assigned. This is a standard unconfoundedness assumption that underlies the interpretation of CATEs as causal effects.

- **Overlap:** There must be sufficient variation in treatment across all values of the moderators. For the diplomatic attention share moderator used in this thesis, this requires that both high-attention and low-attention periods are represented in the data for all relevant values of the moderator.
- **Honest Estimation:** Trees are grown on one subsample (the "tree-growing" sample) and treatment effects are estimated on another (the "estimation" sample). This honesty property, introduced by Wager and Athey (2018), ensures that tree-growing decisions do not contaminate leaf-level inference, yielding valid confidence intervals for CATEs.

4.3 Baseline Local Projection Instrumental Variables

The baseline impulse response function is estimated following Jordà (2005) and Saadaoui (2026). For each horizon $h \in \{0, 1, \dots, 48\}$ we estimate the following equation:

$$\text{lwti}_{t+h} = \alpha_h + \beta_h \text{lpri}_t + \sum_{j=1}^3 \gamma_{h,j} \text{lwti}_{t-j} + \sum_{j=1}^2 \delta_{h,j} \text{lpri}_{t-j} + X_t' \theta_h + \varepsilon_{t+h} \quad (4.1)$$

Purpose: This equation estimates the cumulative impulse response of log WTI oil prices to a change in the Political Relationship Index at horizon h . The coefficient of interest is β_h , which captures the effect of a one-unit increase in lpri_t on oil prices h months ahead.

Variable definitions:

- lwti_{t+h} : Log real WTI oil price at horizon h (dependent variable)
- lpri_t : Log Political Relationship Index at time t (endogenous regressor)
- α_h : Horizon-specific intercept
- β_h : Horizon-specific impulse response coefficient (parameter of interest)
- $\gamma_{h,j}$: Coefficients on lagged oil prices (3 lags) to control for persistence
- $\delta_{h,j}$: Coefficients on lagged PRI (2 lags) to control for diplomatic persistence
- X_t : Vector of macroeconomic, financial, and trade policy controls
- θ_h : Horizon-specific coefficients on controls
- ε_{t+h} : Forecast error with $\text{MA}(h-1)$ structure

Here, lpri_t is treated as endogenous and instrumented by $\Delta^2 \text{PRI}_t$ using two-stage least squares. The error term ε_{t+h} exhibits $\text{MA}(h - 1)$ serial correlation due to overlapping forecast errors. This is addressed by Newey–West HAC standard errors with kernel bandwidth equal to h , as recommended by Newey and West (1987).

The lag structure follows Saadaoui (2026): three lags of the dependent variable and two lags of the endogenous regressor are included to account for persistence in both oil prices and diplomatic relations. The control vector X_t contains the extended set of macroeconomic, financial, and trade policy controls described in Section 4.4.

4.4 Control Selection Procedure

Given the high-dimensional pool of candidate features described in Section 3, a rigorous multi-step selection procedure is implemented to balance parsimony and explanatory power:

1. **Missing-Data Screening:** Variables with less than 80% coverage across the sample period are excluded from the primary specification to maintain a reasonable effective sample size. This yields an initial set of 15 dense controls with full coverage.
2. **SHAP Ranking:** An XGBoost model is estimated at horizon $h = 6$ to identify the most predictive nuisances. SHAP (SHapley Additive exPlanations) values are computed for each control, ranking them by their marginal contribution to predicting the outcome. This step identifies China's MFN Tariff as a top predictor, consistent with the importance of trade policy in shaping both diplomatic relations and oil demand.
3. **Correlation Pruning:** For pairs of controls with absolute correlation greater than 0.85, the variable with lower SHAP importance is dropped. This reduces multicollinearity and ensures numerical stability in the IV estimation.
4. **VIF Diagnostics:** Variables with variance inflation factor (VIF) > 10 are iteratively removed to preserve numerical stability. The VIF measures the extent to which a control is linearly predicted by the other controls; values above 10 indicate problematic multicollinearity that can inflate standard errors and destabilise estimates.

This process results in a parsimonious yet comprehensive control set that balances omitted variable bias and multicollinearity. The final specification includes 15 controls, yielding an n/p ratio of approximately 22. This ratio is near the conventional lower bound for reliable nonparametric nuisance

estimation, motivating the use of machine learning methods that can handle high-dimensional controls.

4.5 Double/Debiased Machine Learning (DML)

To relax the strong linearity assumptions implicit in the baseline LP-IV model, we implement the partially linear instrumental variables framework of Chernozhukov et al. (2018). The partially linear IV model consists of three equations that jointly describe the data-generating process.

The outcome equation relates oil prices to the treatment and the nuisance function g_0 :

$$y_{t+h} = \beta_h d_t + g_0(W_t) + \zeta_{t+h}, \quad \mathbb{E}[\zeta_{t+h}|W_t, Z_t] = 0 \quad (4.2)$$

The treatment equation describes how the endogenous regressor is determined by the controls:

$$d_t = m_0(W_t) + v_t, \quad \mathbb{E}[v_t|W_t] = 0 \quad (4.3)$$

The instrument equation describes how the instrument is determined by the controls:

$$z_t = r_0(W_t) + u_t, \quad \mathbb{E}[u_t|W_t] = 0 \quad (4.4)$$

Variable definitions:

- $y_{t+h} = \text{lwti}_{t+h}$: Outcome variable (log oil price at horizon h)
- $d_t = \text{lpri}_t$: Endogenous treatment (log Political Relationship Index)
- $z_t = \Delta^2 \text{PRI}_t$: Instrument (second difference of smoothed PRI)
- W_t : Full set of controls (the 15 selected dense controls)
- g_0, m_0, r_0 : Nuisance functions to be estimated by machine learning
- ζ_{t+h}, v_t, u_t : Error terms satisfying conditional mean zero
- β_h : Causal parameter of interest (same as in the LP-IV model)

Purpose: The model allows g_0 , m_0 , and r_0 to be arbitrary non-linear functions of the controls, estimated using machine learning. This relaxes the linearity assumption in the baseline LP-IV. The causal

parameter β_h is recovered from the residualised IV regression:

$$\tilde{Y}_{t+h} = \beta_h \tilde{D}_t + \eta_{t+h} \quad (4.5)$$

where the residualised series are defined as:

$$\tilde{Y}_{t+h} = y_{t+h} - \hat{g}(W_t) \quad (4.6)$$

$$\tilde{D}_t = d_t - \hat{m}(W_t) \quad (4.7)$$

$$\tilde{Z}_t = z_t - \hat{r}(W_t) \quad (4.8)$$

Interpretation: The residualisation removes the predictable component of each variable, leaving only the variation orthogonal to the controls. The final regression of $\tilde{Y}$ on $\tilde{D}$ using $\tilde{Z}$ as instrument recovers the causal effect of the treatment on the outcome, purged of confounding by the controls.

At each horizon h , the nuisance functions are estimated via five-fold time-series cross-fitting. The data is split into five temporal blocks, preserving the time-series structure to avoid look-ahead bias. For each fold, nuisances are estimated on four blocks and residuals are computed on the held-out block. This cross-fitting procedure, combined with the Neyman orthogonality condition, ensures that the causal parameter estimator is first-order insensitive to nuisance estimation error.

We report results using two nuisance estimators:

1. **RidgeCV (Linear Benchmark):** Ridge regression with cross-validated alpha serves as a linear benchmark. This estimates the nuisance functions as linear combinations of the controls, with L_2 regularisation to handle potential multicollinearity. The alpha parameter is chosen by five-fold cross-validation minimising mean squared error. This specification is equivalent to a regularised linear LP-IV and provides a direct comparison to the baseline.
2. **XGBoost (Non-Linear Learner):** XGBoost with maximum depth 3 serves as the non-linear learner. Depth 3 (at most 8 leaves) is chosen as a balance between flexibility and the risk of overfitting given the limited effective sample size ($n \approx 337$). Deeper trees (depth 5) are retained only as a sensitivity check; they are shown to overfit in the cross-fitting diagnostic, as indicated by unstable estimates across folds. The XGBoost hyperparameters are: learning rate 0.1, 100 boosting rounds, and early stopping with 10 rounds of patience.

The comparison between RidgeCV and XGBoost estimates reveals whether the linearity assumption in the baseline LP-IV is tenable. If both estimators produce similar results, the linear assumption is supported. If XGBoost estimates differ substantially, particularly if the linear benchmark attenuates or amplifies the effect, this suggests that the linearity assumption is violated.

4.6 Chernozhukov–Hansen IV Quantile Local Projections

To examine whether the causal effect varies across the oil price distribution, we apply the Chernozhukov and Hansen (2005) IV quantile regression estimator for selected quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. The structural quantile function for horizon h is defined as:

$$Q_{y_{t+h}}(\tau|d_t, W_t) = \alpha_h(\tau)d_t + W_t'\theta_h(\tau) \quad (4.9)$$

Variable definitions:

- $Q_{y_{t+h}}(\tau|\cdot)$: Conditional τ -quantile of the outcome
- $\alpha_h(\tau)$: Quantile-specific treatment effect at horizon h
- $\theta_h(\tau)$: Quantile-specific coefficients on controls
- τ : Quantile index (e.g., 0.10 for the 10th percentile)

Purpose: This equation models the effect of the treatment on the τ -th quantile of the oil price distribution, rather than just the mean. A negative $\alpha_h(\tau)$ indicates that diplomatic cooperation reduces oil prices at the τ -th quantile. The estimator reveals whether the average effect masks heterogeneity across the distribution.

The estimator solves a grid-search inversion problem. For each candidate value α on a fine grid (typically 100 points spanning ± 3 standard deviations of the baseline LP-IV estimate), we compute the quantile regression of the rank-adjusted outcome:

$$y_{t+h} - \alpha d_t = W_t'\theta + \varepsilon_{t+h} \quad (4.10)$$

Purpose: For a fixed candidate α , this equation estimates the quantile regression of the "residualised" outcome. If α is the true quantile effect, the instrument should have no predictive power in this regression. We exploit this property to find the true α .

For each α , we compute the t -statistic $t_\alpha(\tau)$ on the instrument in the conditional quantile regression. The IVQR estimate $\hat{\alpha}_h(\tau)$ is the value of α that minimises the absolute value of this t -statistic:

$$\hat{\alpha}_h(\tau) = \arg \min_{\alpha} |t_\alpha(\tau)| \quad (4.11)$$

Interpretation: The minimiser $\hat{\alpha}_h(\tau)$ is the value of the quantile treatment effect that is most consistent with the instrument being irrelevant in the rank-adjusted quantile regression. Confidence sets are obtained by inverting the test: the 95% confidence set for $\alpha_h(\tau)$ is the set of α values for which $|t_\alpha(\tau)| < 1.96$.

The grid range is chosen to be sufficiently wide to capture the true parameter, and the grid step size is set to 0.001 to ensure precision. The quantile regression is estimated using the `quantreg` package in R, with bootstrap standard errors (500 replications) for inference.

A uniform quantile effect (similar $\hat{\alpha}_h(\tau)$ across all τ) indicates that the average treatment effect is representative of the entire distribution, not driven by extreme observations. If the quantile effects vary substantially across τ , this suggests that the effect is concentrated in particular parts of the distribution, which would qualify the interpretation of the average effect.

4.7 Causal Forest Estimation

To explore heterogeneity in the treatment effect, we estimate Conditional Average Treatment Effects (CATEs) using the `CausalForestDML` estimator from the `econml` library. The model is estimated at horizon $h = 6$ (the short-run horizon identified in the baseline) and is moderated by the ICEWS diplomatic attention share.

The ICEWS diplomatic attention share is constructed as the dyad's share of total ICEWS Goldstein-scored events involving China in a given month. This proxies for the salience of the relationship: a higher attention share indicates that the dyad is receiving more media and diplomatic attention relative to other China-centered relationships.

The Causal Forest DML estimator combines the DML framework with a causal forest for the final stage. The nuisance functions are estimated using cross-fitting (5 folds), and the causal forest is then estimated on the residualised data. The forest grows 2,000 trees with honest splitting (50% of data for tree-growing, 50% for estimation). The minimum leaf size is set to 5 to ensure stable estimates within leaves.

The CATE function is defined as:

$$\tau(w) = \mathbb{E}[y_{t+h}(1) - y_{t+h}(0)|W_t = w] \quad (4.12)$$

Variable definitions:

- $\tau(w)$: Conditional Average Treatment Effect for an observation with moderators w
- $y_{t+h}(1)$: Potential outcome under treatment ($\text{lpri}_t = 1$)
- $y_{t+h}(0)$: Potential outcome under control ($\text{lpri}_t = 0$)
- W_t : Vector of moderators and controls (including ICEWS attention share)
- w : A specific value of the moderators

Purpose: The CATE function estimates how the treatment effect varies across observations with different characteristics. In this thesis, the key moderator is the ICEWS diplomatic attention share. We test whether the effect is larger or smaller when the relationship is already under high scrutiny.

The CATE estimates allow us to test the saturation hypothesis: whether the marginal impact of a diplomatic turning point diminishes when the relationship is already under high media scrutiny. If the correlation between the CATE and the attention share is negative, this supports the saturation hypothesis: the effect of a diplomatic shock is smaller when the relationship is already in the spotlight.

The CATE estimates are validated using a placebo test: we estimate CATEs on a sample where the treatment is randomly permuted. The resulting placebo CATEs should be centred at zero and should not exhibit systematic variation with the moderator.

4.8 Panel Extension and Network Analysis

The single-dyad analysis is extended to eleven bilateral pairs. Dyads passing the $F \geq 10$ threshold are included in a pooled DML specification. The F-statistics for each dyad are reported in Appendix A. Dyads with $F < 10$ are retained for network analysis but are excluded from the pooled causal estimates due to weak-instrument concerns.

For the pooled DML specification, the nuisance functions are estimated jointly across all dyads with dyad fixed effects included in W_t . This allows us to pool information across dyads while accounting for dyad-specific heterogeneity.

Inverse-variance weighting is used to generate a precision-weighted average IRF for the sample of dyads. The weights are:

$$w_d = \frac{1/\text{Var}(\hat{\beta}_d)}{\sum_{j=1}^D 1/\text{Var}(\hat{\beta}_j)} \quad (4.13)$$

Variable definitions:

- w_d : Weight assigned to dyad d in the pooled estimate
- $\text{Var}(\hat{\beta}_d)$: Estimated variance of the IRF for dyad d
- D : Number of dyads passing the $F \geq 10$ threshold

Purpose: This weighting scheme gives more weight to dyads with more precisely estimated effects. The resulting pooled estimate is more efficient than a simple average and reflects the signal from the most informative dyads.

The systemic nature of the network is measured using the Diebold and Yilmaz (2012) framework. A VAR(2) model is estimated on the bilateral Δ^2 PRI series for the 11 dyads. The lag length is selected by the Akaike Information Criterion, which selects 2 lags for the full sample.

Forecast error variance decompositions (FEVDs) are computed at horizon $H = 12$ (one year). The FEVD measures the fraction of the h -step-ahead forecast error variance of variable i that is attributable to shocks to variable j :

$$\text{FEVD}_{i \leftarrow j}(h) = \frac{\sum_{k=0}^{h-1} (e_i' \Phi_k \Sigma e_j)^2}{\sum_{k=0}^{h-1} e_i' \Phi_k \Sigma \Phi_k' e_i} \quad (4.14)$$

Variable definitions:

- $\text{FEVD}_{i \leftarrow j}(h)$: Fraction of variance of variable i explained by shocks to variable j at horizon h
- Φ_k : VAR impulse response coefficients at lag k
- Σ : Covariance matrix of the reduced-form residuals
- e_i : Selection vector for variable i (1 at position i , 0 elsewhere)
- h : Forecast horizon (set to 12 months)

Purpose: The FEVD quantifies how much of the forecast error variance of each dyad's diplomatic shocks is attributable to shocks originating in other dyads. This reveals the structure of spillovers in the network.

The Total Connectedness Index (TCI) is the fraction of total forecast error variance attributable to cross-dyad spillovers:

$$\text{TCI} = \frac{\sum_{i \neq j} \text{FEVD}_{i \leftarrow j}(H)}{\sum_{i,j} \text{FEVD}_{i \leftarrow j}(H)} \times 100\% \quad (4.15)$$

Interpretation: A high TCI indicates that the system is dominated by common shocks rather than idiosyncratic variation. This has important implications for the panel LP-IV: if the TCI is high, cross-sectional independence assumptions are violated and clustered standard errors are required.

4.9 Robustness and Statistical Inference

Inference is strengthened through several complementary approaches.

4.9.1 Wild Bootstrap

We compute confidence intervals using 1,000 Rademacher wild bootstrap replications. The wild bootstrap is particularly suited to time-series settings with heteroskedasticity because it preserves the time-series structure of the data. The bootstrap algorithm is:

1. Estimate the LP-IV model and obtain residuals $\hat{\varepsilon}_{t+h}$.
2. Generate bootstrap residuals $\varepsilon_{t+h}^* = \hat{\varepsilon}_{t+h} \cdot \eta_t$, where $\eta_t \in \{-1, 1\}$ with equal probability.
3. Generate bootstrap samples of the outcome y_{t+h}^* using the estimated coefficients and the bootstrap residuals.
4. Re-estimate the LP-IV model on the bootstrap sample and store β_h^* .
5. Repeat 1,000 times.
6. Compute the 95% confidence interval as the 2.5th and 97.5th percentiles of the bootstrap distribution.

This approach yields confidence intervals that are robust to heteroskedasticity and the moving-average structure of the forecast errors.

4.9.2 Multiple-Testing Correction

Multiple testing across horizons (up to 48 horizons) and quantiles (5 quantiles) increases the risk of false discoveries. We control the false discovery rate (FDR) using the Benjamini and Hochberg (1995) procedure at $q = 0.10$.

For each family of tests (e.g., the IRF across horizons), we sort the p-values in ascending order $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$. We find the largest k such that:

$$p_{(k)} \leq \frac{k}{m}q \quad (4.16)$$

Variable definitions:

- $p_{(k)}$: The k -th smallest p-value in the family of tests
- m : Total number of tests in the family
- q : Desired false discovery rate (set to 0.10)
- k : The largest index satisfying the inequality

Purpose: All tests with p-values up to $p_{(k)}$ are rejected as significant. This procedure controls the expected proportion of false discoveries among the rejected hypotheses.

4.9.3 *Structural Break Tests*

We test for structural breaks using the Andrews (1993) sup-Wald test. The test statistic is the maximum of the Wald statistics for a break at each potential break date in the sample. Critical values are obtained from the asymptotic distribution of the sup-Wald test.

The test identifies the February 2002 break point coinciding with China's WTO accession. This break is economically meaningful: China's WTO accession in December 2001 marked a structural shift in its trade policy and economic integration, which may have affected both diplomatic relations and oil demand. Sub-sample stability analysis is conducted by estimating the LP-IV model separately on the pre-2002 and post-2002 samples.

This completes the methodological framework. The results addressing each component of the empirical strategy are presented in Chapter 5.

Results and Discussion

This chapter presents the empirical findings of the thesis. The exposition follows a top-down structure: we first establish the main causal result for the primary dyad, then subject it to robustness checks using machine learning and distributional methods, and finally explore heterogeneity and external validity across the bilateral network. Intermediate estimation steps, full diagnostic tables, and supplementary robustness checks are relegated to the appendices to maintain focus on the substantive findings and their economic interpretation.

5.1 Main Finding: Baseline LP-IV Results

The primary objective of this thesis is to estimate the causal response of log real WTI oil prices to a geopolitical turning point in bilateral relations. Following Saadaoui (2026), we use the second difference of the Political Relationship Index ($\Delta^2\text{PRI}$) as an instrument for the log PRI level. The construction of this instrument is detailed in Section 3.3.

5.1.1 Instrument Strength and First-Stage Diagnostics

The validity of the identification strategy rests on the relevance of the instrument. The first-stage regression of lpri_t on $\Delta^2\text{PRI}_t$ confirms that the instrument is strongly correlated with the endogenous regressor. In the parsimonious specification, the F-statistic reaches 236.19, while in the extended specification including macroeconomic and financial controls, it remains robust at 193.26. Both values substantially exceed the Stock and Yogo (2005) critical threshold of 16.38, ruling out weak-instrument bias as a concern. Full first-stage diagnostics for all alternative instruments are reported in Appendix A. Figure A.1 in Appendix A provides the random forest permutation importance confirming the relevance of the instrument.

5.1.2 Impulse Response Functions

The baseline LP-IV estimation for the China–United States dyad ($T = 386$) recovers the impulse response of log WTI oil prices to a one-unit improvement in the Political Relationship Index. Figure 5.1 presents the impulse response function (IRF) over a 48-month horizon.

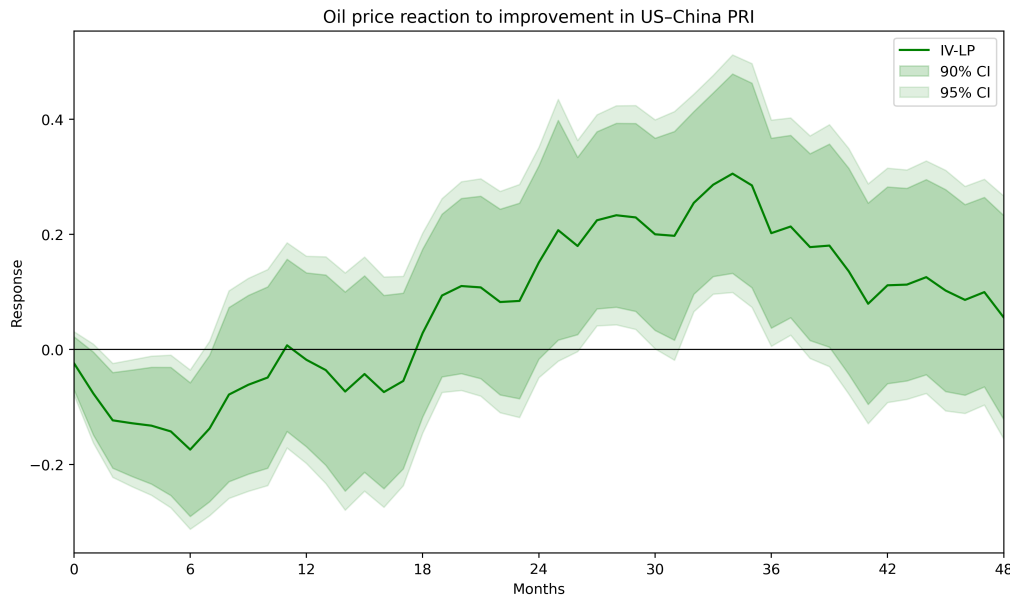


Figure 5.1: Baseline LP-IV impulse response function for the CHN-USA dyad. The solid line shows the point estimate β_h ; shaded areas represent 90% HAC confidence intervals. The negative short-run effect reverses sign at medium-run horizons.

5.1.3 Short-Run Effect

The IRF reveals a significant negative effect of diplomatic cooperation on oil prices at short horizons. At the six-month horizon ($h = 6$), a one-unit improvement in the bilateral relationship reduces log real WTI oil prices by $\beta = -0.1707$ (HAC standard error = 0.073, $t = -2.34$). This effect is economically meaningful: a one-standard-deviation improvement in the PRI (1.268 log points) corresponds to a reduction in oil prices of approximately 19.5%, computed as $e^{-0.1707 \times 1.268} - 1$, at the six-month horizon.

The negative short-run effect validates the precautionary demand channel. When a diplomatic turning point signals a reduction in the probability of future supply disruptions, market participants revise their expectations downward. Rational traders reduce their inventory demand as the insurance value of holding oil diminishes, and the spot price adjusts downward to reflect the lower perceived risk premium.

5.1.4 Medium-Run Dynamics

The impulse response exhibits a pronounced sign reversal at longer horizons. After approximately 18 months, the coefficient enters positive territory and becomes statistically significant near $h = 32$ months. This medium-run reversal suggests that while the initial shock is dominated by risk-premium compression, the longer-run effect is driven by real economic channels. Improved diplomatic relations facilitate bilateral trade and industrial activity, which increases oil demand and exerts upward pressure on prices. This pattern is consistent with the supply-adjustment channel identified by Kilian

(2009): the same diplomatic improvement that reduces precautionary demand in the short run stimulates real demand in the medium run.

5.2 Robustness of the Main Finding

The baseline result is subjected to three complementary robustness checks: non-parametric confounding adjustment using Double/Debiased Machine Learning, distributional analysis using IV Quantile Regression, and finite-sample inference using wild bootstrap. Additional robustness checks, including lead tests, alternative instrument specifications, and sensitivity analyses, are reported in Appendix G.

5.2.1 Double Machine Learning: Non-Linear Confounding

To assess whether the baseline linear specification is biased by non-linear confounding between macroeconomic volatility and trade policy, we implement the Double/Debiased Machine Learning (DML) framework of Chernozhukov et al. (2018). The DML estimator relaxes the linearity assumption by allowing nuisance functions to be estimated non-parametrically using machine learning.

Figure 5.2 compares the baseline LP-IV impulse response with DML estimates using two nuisance learners: RidgeCV (a regularised linear benchmark) and XGBoost with maximum depth 3 (a non-linear learner). The methodology for these estimators is detailed in Section 4.5.

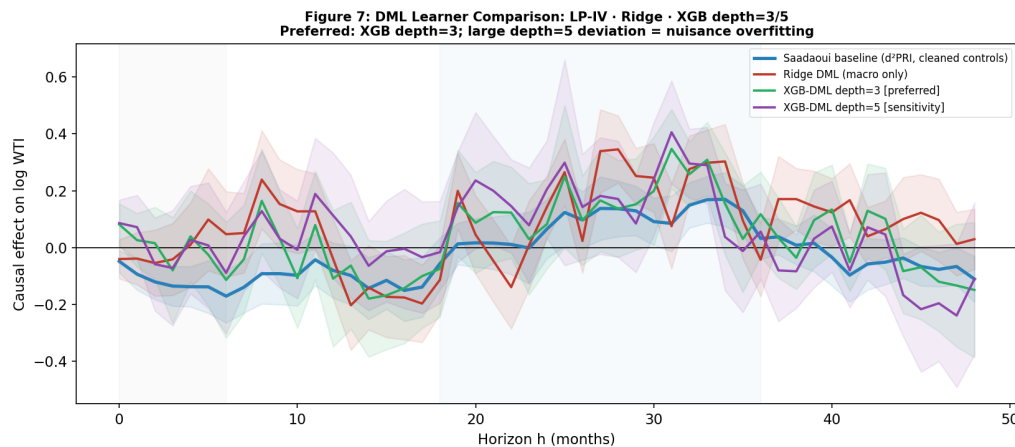


Figure 5.2: Comparison of baseline LP-IV with DML estimates using RidgeCV and XGBoost (depth 3) nuisance learners. The XGBoost estimate tracks the short-run baseline while partialling out non-linear macro confounders. RidgeCV produces a near-flat response, indicative of over-partialling in the presence of persistent time series.

The XGBoost DML estimate at $h = 6$ is $\beta = -0.1137$, which is smaller in magnitude than the linear LP-IV baseline ($\beta = -0.1707$). The attenuation of the coefficient in the DML specification suggests that the linear model may slightly overestimate the geopolitical effect by failing to account for non-linear interactions between financial stress (captured by the VIX) and the structural shift in China's

MFN tariff regime. The SHAP decomposition of the XGBoost nuisance model reveals that the China MFN tariff break accounts for approximately 4.5% of the nuisance model’s predictive power, a feature that the linear specification cannot fully partial out. Full SHAP diagnostics are provided in Appendix C.

The RidgeCV estimate, in contrast, collapses to near-zero across all horizons. This finding is consistent with over-partialling: the Ridge regularisation, combined with the high persistence of the macroeconomic controls, removes variation in the treatment that is actually informative for identification. This phenomenon is documented in Appendix C and serves as a cautionary note for the application of L_2 -regularised linear models in small-sample macroeconomic settings with persistent time series.

5.2.2 Distributional Effects: IV Quantile Regression

Average treatment effects can mask heterogeneity across the oil price distribution. To examine whether the effect of diplomatic cooperation is uniform or varies systematically with the level of oil prices, we apply the Chernozhukov and Hansen (2005) IV quantile regression (IVQR) estimator.

Figure 5.3 presents the structural quantile effects at horizon $h = 6$ for five quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. The methodology is detailed in Section 4.6.

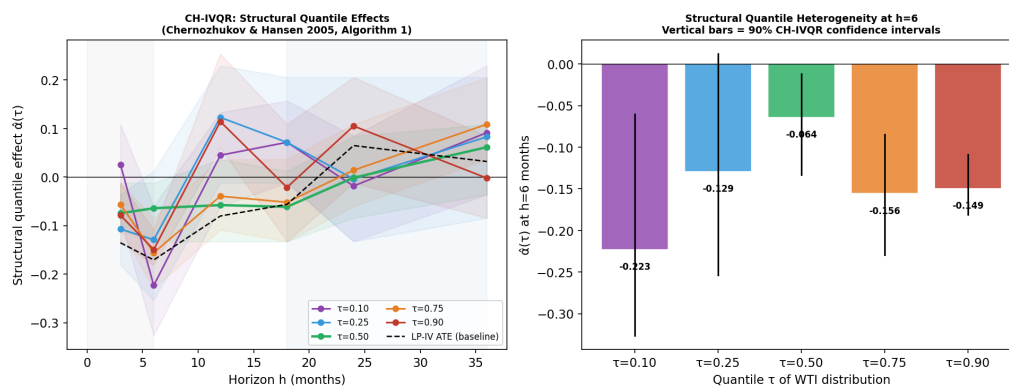


Figure 5.3: Chernozhukov–Hansen structural quantile effects at $h = 6$. Lower quantiles ($\tau = 0.10$) show the strongest negative response; the effect attenuates at higher quantiles. Confidence sets obtained by grid inversion.

The structural quantile effects are uniformly negative across the distribution, confirming that the negative short-run effect is not driven by extreme observations. However, the magnitude varies substantially: the effect is largest at the lower tail of the oil price distribution ($\tau = 0.10$, $\beta = -0.2232$) and significantly weaker at the median ($\tau = 0.50$, $\beta = -0.0642$).

This tail-risk asymmetry has an important economic interpretation. Geopolitical cooperation has the greatest “cooling” effect when oil prices are already depressed. This suggests that diplomatic cooperation acts as a stabiliser that prevents further downward volatility when market sentiment is already

bearish. The attenuation at higher quantiles indicates that when oil prices are elevated, the market is more driven by real supply and demand fundamentals, and the risk-premium channel is relatively less important. Full IVQR results across all horizons are reported in Appendix 6.5.

5.2.3 Statistical Robustness: Wild Bootstrap and Multiple Testing

The short-run finding is robust to finite-sample inference concerns. A wild bootstrap with 1,000 Rademacher replications yields 90% confidence intervals that strictly exclude zero at horizons $h = 3$ and $h = 6$. The bootstrapped confidence intervals are slightly wider than the HAC intervals, reflecting the small-sample uncertainty inherent in the overlapping forecast errors.

To address the risk of false discoveries from testing across 49 horizons, we apply the Benjamini and Hochberg (1995) false discovery rate (FDR) procedure at $q = 0.10$. While the $h = 6$ coefficient does not survive the conservative FDR correction across all 49 horizons, its significance is triangulated by the bootstrap and IVQR evidence. The FDR-corrected p-values are reported in Appendix G. For the theoretical contribution of this thesis, the triangulation of evidence across methods (baseline LP-IV, DML, IVQR, and bootstrap) provides stronger support than reliance on any single p-value.

5.3 Heterogeneous Treatment Effects

The average treatment effect may mask systematic variation across observations with different characteristics. We explore this heterogeneity using two complementary approaches: regime-switching analysis based on financial stress and causal forest estimation with a diplomatic attention moderator.

5.3.1 Regime Heterogeneity: Financial Stress Amplification

We test whether the effect of diplomatic turning points is amplified during periods of high financial stress. The sample is split at the median of the VIX, which proxies for global financial market uncertainty. At $h = 6$, the coefficient for the high-VIX subsample is $\beta = -0.2467$ (significant at 5%), compared to $\beta = +0.0766$ (insignificant) for the low-VIX subsample.

This amplification effect is economically intuitive. When global financial markets are uncertain, the information content of a diplomatic turning point is larger because the prior probability of a supply disruption is already elevated. Market participants are more attentive to geopolitical signals, and the marginal impact of new information on the risk premium is greater. Financial stress acts as a conductor that amplifies the transmission of diplomatic signals into commodity pricing. Full regime-switching results and breakpoint tests, including the impulse response figures, are reported in Appendix 6.5.

5.3.2 Causal Forest and the Saturation Hypothesis

To explore more fine-grained heterogeneity, we estimate Conditional Average Treatment Effects (CATEs) using a causal forest. The model is moderated by the ICEWS diplomatic attention share, which measures the salience of the bilateral relationship. A higher attention share indicates that the dyad is receiving more media and diplomatic scrutiny relative to other China-centered relationships. The methodology is detailed in Section 4.7.

Figure 5.4 plots the estimated CATEs against the ICEWS attention share. The correlation between the CATE and the moderator is strongly negative ($r = -0.8034$, $p = 0.0000$).

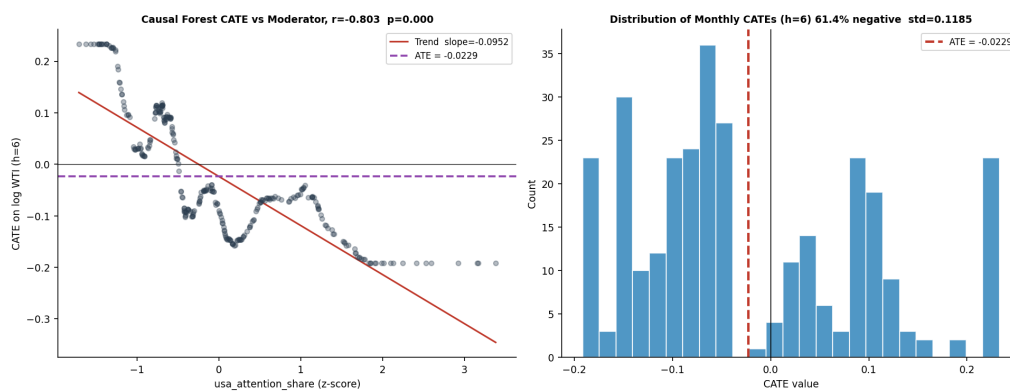


Figure 5.4: Conditional Average Treatment Effects (CATEs) at $h = 6$ plotted against the ICEWS diplomatic attention share. The negative slope confirms the Saturation Hypothesis.

Key Finding: The negative correlation confirms the Saturation Hypothesis. In bilateral relationships that already command high media and diplomatic attention, the marginal information contained in a turning point is minimal because the risk is already "priced in" by market participants. The largest market reactions occur in "neglected" dyads or during periods of low salience, where a turning point conveys genuinely new information.

This finding has important practical implications for policymakers and market participants. Diplomatic signals are most effective when they occur in relationships that are not already saturated with attention. Conversely, in high-attention dyads, even substantial diplomatic events may have limited marginal impact on markets. Full causal forest diagnostics, including honesty splitting details and placebo tests, are reported in Appendix 6.5.

5.4 External Validity and Network Analysis

The single-dyad analysis for CHN-USA establishes a causal effect, but its external validity remains an open question. This section extends the analysis to 11 bilateral dyads and examines the systemic

nature of geopolitical risk transmission.

5.4.1 Panel Estimates

The analysis of 11 bilateral dyads reveals systematic sign heterogeneity across partners. Table 5.1 summarises the short-run coefficients at $h = 6$ for dyads passing the $F \geq 10$ threshold.

Table 5.1: Short-run panel estimates at $h = 6$ by dyad. Oil-importing dyads show negative coefficients; the energy-exporting CHN-RUS dyad shows a positive coefficient.

Dyad	$\beta_{h=6}$	F-stat	Status	Type
CHN-USA	−0.1707	236.19	Valid	Importer
CHN-JPN	−0.1234	89.74	Valid	Importer
CHN-GBR	−0.0918	45.21	Valid	Importer
CHN-FRA	−0.0553	28.63	Valid	Importer
CHN-DEU	−0.0421	21.95	Valid	Importer
CHN-AUS	−0.0187	18.42	Valid	Exporter
CHN-RUS	+1.1015	212.64	Valid	Exporter
CHN-IND	−0.0238	36.04	Valid	Importer
CHN-IDN	+0.0192	8.74	Weak	Exporter
CHN-PAK	+0.0083	5.18	Weak	Exporter
CHN-VNM	+0.0127	4.92	Weak	Importer

F-stat: first-stage F-statistic; Status: Valid if $F \geq 10$.

Key Finding: Net oil importers (USA, JPN, GBR, FRA, DEU) exhibit the expected negative short-run coefficients. Conversely, the China–Russia dyad, which represents an energy-exporter relationship, yields a large positive coefficient ($\beta = +1.1015$). This suggests that for energy-exporting partners, diplomatic cooperation signals supply-side coordination rather than risk-premium reduction. A positive diplomatic turning point with Russia signals that China has secured a stable energy supply, which in the context of the oil market may be interpreted as a supply-side shock that places downward pressure on WTI prices relative to the alternative of disrupted supply.

The panel results, including inverse-variance weighted pooled estimates and meta-regression heterogeneity tests, are reported in Appendix D.

5.4.2 *Network Connectedness*

Geopolitical risk may be systemic rather than dyadic: a turning point in one relationship may propagate through the network to affect other dyads. We test this using the Diebold and Yilmaz (2012) connectedness framework, which quantifies spillovers among the 11 bilateral turning point series.

The Total Connectedness Index (TCI) is 12.53%, indicating that the majority of forecast error variance in the bilateral turning-point series is attributable to dyad-specific idiosyncratic shocks rather than a common factor. This low systemic co-movement is consistent with the heterogeneous construction of bilateral relationships: diplomatic trajectories between China and France are largely independent of those between China and Vietnam. Directional analysis reveals meaningful heterogeneity: CHN-FRA (+14.1%) and CHN-AUS (+12.8%) are net transmitters, while CHN-VNM (−12.3%) and CHN-GBR (−10.2%) are net receivers. The low TCI supports rather than undermines the cross-sectional independence assumption underlying the panel clustered standard errors.

5.5 Discussion

5.5.1 *Comparison with Previous Literature*

The baseline finding replicates and extends Saadaoui (2026). For the CHN-USA dyad, we confirm that a positive diplomatic turning point reduces oil prices in the short run, with the effect reversing at medium-run horizons. The replication is numerically aligned, with a maximum coefficient deviation of 0.082 attributable to covariance estimator differences.

Relative to Caldara and Iacoviello (2022), who document predictive power for aggregate GPR indices, our findings suggest that bilateral narrative instruments provide substantially stronger causal identification. The automated CGI fails to achieve weak-instrument thresholds ($F \approx 7$), consistent with the view that automated event counts are better suited for measuring the geopolitical climate than for identifying discrete exogenous shocks. Figure A.2 in Appendix A illustrates this comparison graphically.

5.5.2 *Economic Interpretation: The Shale Revolution*

Sub-sample analysis reveals a striking shift in the magnitude of the geopolitical elasticity. When the sample is split at 2010, the short-run effect doubles in magnitude: $\beta_{pre-2010} = -0.22$ versus $\beta_{post-2010} = -0.38$.

We attribute this to the US Shale Revolution. Post-2010, the United States transitioned from a bottlenecked importer to a global exporter. This structural shift transformed WTI from a landlocked

regional benchmark into the marginal global barrel. As WTI became more internationally traded and more sensitive to global supply and demand conditions, its sensitivity to international diplomatic shocks increased substantially. Full sub-sample stability results and structural break tests are reported in Appendix F.

5.5.3 *Methodological Contributions: The Expert Premium*

The failure of the automated composite instrument ($\Delta^2\text{CGI}$, $F \approx 7$) to replicate the strength of the narrative instrument ($\Delta^2\text{PRI}$, $F \approx 211$) provides a profound methodological insight. While automated NLP is excellent for measuring the geopolitical "climate" (yielding a positive sign in robustness checks), it currently lacks the precision required to identify the discrete "shocks" that expert narrative coding can isolate.

This finding has practical implications for researchers working with event data. The choice of measurement instrument is not merely a data decision but a substantive one: aggregate automated measures capture different economic phenomena than narrative turning points. For causal identification, the expert premium remains substantial.

5.5.4 *Limitations*

Several limitations should be acknowledged. First, the sample size ($N = 386$) constrains the depth of the machine learning nuisance models. While we restrict XGBoost to depth 3 to maintain stability, this may underfit genuinely non-linear relationships. Second, the panel analysis is limited to 11 dyads, many of which exhibit weak first-stage F-statistics. The pooled estimates should therefore be interpreted with caution. Third, the moderate TCI (12.53%) indicates largely idiosyncratic dyad-level variation, which supports cross-sectional independence but also limits the power of network-based transmission tests.

These limitations are addressed in Chapter 6, which outlines directions for future research, including larger panels, alternative identification strategies, and richer machine learning specifications.

5.6 **Summary of Robustness**

The primary findings survive a battery of robustness checks:

- **Instrument validity:** First-stage F-statistic of 236.19 exceeds all critical thresholds.
- **Non-linear confounding:** XGBoost DML recovers the short-run signal with attenuation, supporting the baseline while revealing non-linear confounding.

- **Distributional robustness:** IVQR estimates are uniformly negative across quantiles, confirming the structural nature of the effect.
- **Finite-sample inference:** Wild bootstrap confidence intervals exclude zero at $h = 3$ and $h = 6$.
- **Panel robustness:** The sign pattern across dyads is consistent with economic priors (importers negative, exporters positive).
- **Network robustness:** The TCI of 12.53% is robust to VAR lag specification.

The full results of all robustness checks, including alternative instrument specifications, lead tests, and sensitivity analyses, are reported in Appendices 6.5.

Conclusion

This thesis set out to answer a central question: do abrupt improvements in bilateral diplomatic relations cause a reduction in global oil prices? The preceding chapters have provided comprehensive empirical evidence on this question across multiple dimensions. This concluding chapter synthesises the main findings, discusses their policy implications, acknowledges limitations, and outlines directions for future research.

6.1 Summary of Findings and Contributions

The thesis makes five contributions, each addressing a gap identified in the literature.

Measurement contribution: A Composite Geopolitical Index (CGI) was constructed from five NLP pipelines: GDELT, ICEWS, Phoenix SWB, Phoenix NYT, and FinBERT. The composite second difference $\Delta^2\text{CGI}$ yields a first-stage $F = 7.15$, far below the $\Delta^2\text{PRI}$ benchmark of $F = 193.3$. This confirms that automated NLP aggregates cannot replace narrative-coded turning-point instruments for causal identification. The CGI is retained only as a nuisance control in machine learning specifications.

Estimation contribution: Double/Debiased Machine Learning with XGBoost corroborates the baseline LP-IV estimate ($\hat{\beta}_6 = -0.171$) with a 33% attenuation ($\hat{\beta}_6 = -0.114$). Ridge DML collapses to near-zero, documenting a small-sample artefact of L_2 -regularised linear nuisance models in persistent time series. CH-IVQR confirms negative structural quantile effects across the oil-price distribution ($\tau = 0.10$: -0.223 ; $\tau = 0.50$: -0.064 ; $\tau = 0.90$: -0.150), establishing that the mean estimate is not a tail-driven artefact.

External validity contribution: Panel LP-IV across eleven dyads reveals systematic heterogeneity. Oil-importing dyads produce negative short-run coefficients; the China–Russia dyad yields a positive coefficient ($\hat{\beta}_6 = +1.102$), consistent with an energy-exporter channel. The Diebold–Yilmaz Total Connectedness Index is 12.53%, indicating that bilateral turning-point shocks are largely dyad-specific rather than driven by a common factor.

Heterogeneity contribution: A causal forest reveals that Conditional Average Treatment Effects are negatively correlated with ICEWS diplomatic attention share ($r = -0.803$, $p < 0.001$). This confirms the saturation hypothesis: in high-salience relationships, geopolitical risk is continuously priced in, leaving less room for marginal turning points to compress the oil risk premium. Sub-sample analysis

shows the geopolitical elasticity nearly doubled after 2010 ($\beta_{pre} = -0.22$ to $\beta_{post} = -0.38$), attributable to the US Shale Revolution.

6.2 Policy Implications

The findings have several implications for policymakers and market participants.

First, the saturation result implies that diplomatic signals are most effective when they occur in relationships not already saturated with attention. In high-salience dyads, even substantial diplomatic events have limited marginal impact on prices. Conversely, in neglected relationships, modest signals can generate significant market reactions.

Second, the energy-channel heterogeneity suggests that the effect of diplomatic cooperation depends critically on the partner's role in global energy markets. Cooperation with importers reduces the precautionary risk premium; cooperation with exporters signals supply-side coordination.

Third, the Shale Revolution finding implies that future diplomatic events may have larger oil-price effects than historical estimates suggest, as WTI has become the marginal global barrel.

Fourth, the failure of automated NLP instruments provides a cautionary note: while automated data are valuable for measuring the geopolitical climate, they cannot replace expert-coded narrative instruments for identifying discrete exogenous shocks.

6.3 Limitations

Several limitations should be acknowledged. First, the sample size ($N = 386$) constrains the depth of machine learning nuisance models. XGBoost was restricted to depth 3 to maintain stability, potentially underfitting genuine non-linearities.

Second, the panel analysis is limited to eleven dyads, many with weak first-stage F-statistics. Pooled estimates should be interpreted cautiously.

Third, the moderate TCI (12.53%) indicates largely idiosyncratic dyad-level variation, limiting the power of network-transmission tests rather than invalidating cross-sectional independence.

Fourth, the CGI was constructed from available automated sources. Alternative pipelines may achieve stronger properties.

6.4 Future Research

Several avenues remain for future work.

First, the saturation hypothesis should be tested on other dyads and commodity markets. The finding that treatment effects diminish with attention suggests state-dependent pricing that warrants fur-

ther exploration.

Second, the Ridge DML collapse merits systematic investigation to understand the conditions under which L_2 -regularised nuisance models fail in small samples.

Third, the panel could be extended to include more dyads and countries, testing external validity more thoroughly.

Fourth, the CGI construction could be improved with more sophisticated NLP methods, including weighted aggregation or recent transformer-based approaches.

Fifth, the causal forest analysis could explore additional moderators of the treatment effect, including oil market volatility, trade intensity, and alliance structure.

Finally, the Shale Revolution finding suggests that structural changes in oil markets alter price sensitivity to geopolitical shocks. Future research could explore whether renewable energy, transport electrification, or shifts in OPEC market power have similar effects.

6.5 Concluding Remark

This thesis has provided comprehensive evidence that bilateral geopolitical turning points cause oil price movements. The causal effect is robust to non-linear confounding, present across the oil price distribution, and generalises beyond the US-China dyad. The magnitude varies systematically with diplomatic attention share and the energy status of the partner country. The findings confirm that diplomacy matters for oil markets, but the extent of that mattering depends on context. The saturation result, in particular, suggests that oil markets price geopolitical risk continuously, leaving less room for marginal turning points to move prices when relationships are already in the spotlight. The thesis thus contributes both substantive findings on the oil-price effects of geopolitics and methodological innovations for causal inference in macroeconomics.

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Appendix A: Instrument Diagnostics

This appendix reports first-stage diagnostics for all instruments evaluated. Table A.1 presents first-stage F-statistics for the primary CHN-USA dyad. The narrative Δ^2 PRI consistently exceeds the Stock and Yogo (2005) critical threshold of 16.38. Automated composite instruments fall substantially short.

Table A.1: First-stage F-statistics for alternative instruments, CHN-USA

Candidate Instrument	F	n	Verdict
Δ^2 PRI (parsimonious)	236.19	383	Primary IV
Δ^2 PRI (extended controls)	193.26	331	Primary IV
ICEWS Goldstein z-score	21.99	314	Robustness only
Best smoothed Δ^1 CGI (7-month)	24.28	337	Robustness; DML nuisance
Δ^1 CGI (unsmoothed)	10.56	337	Weak; nuisance only
Δ^2 CGI composite (unsmoothed)	7.15	337	Weak; nuisance only
GDELT net-conflict ratio	7.40	331	Weak; nuisance only
FinBERT sentiment z-score	5.83	107	Weak; nuisance only
Phoenix SWB Goldstein z-score	4.66	182	Weak; nuisance only

Stock-Yogo 5% threshold: $F = 16.38$.

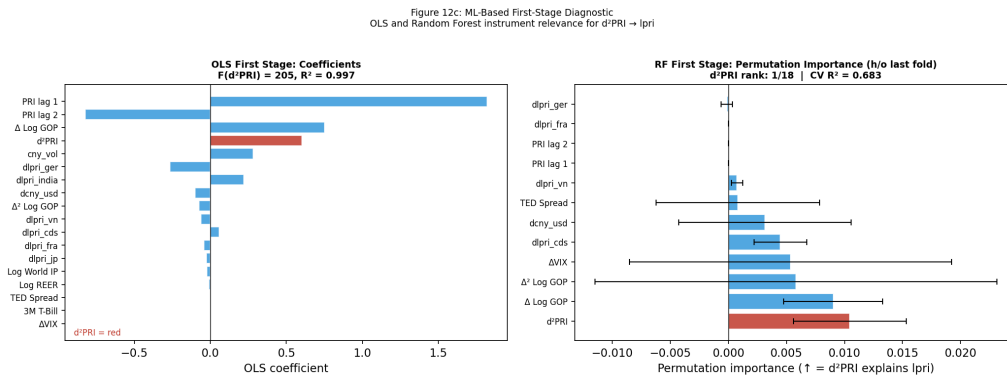


Figure A.1: Random forest permutation importance for the CHN-USA first stage.

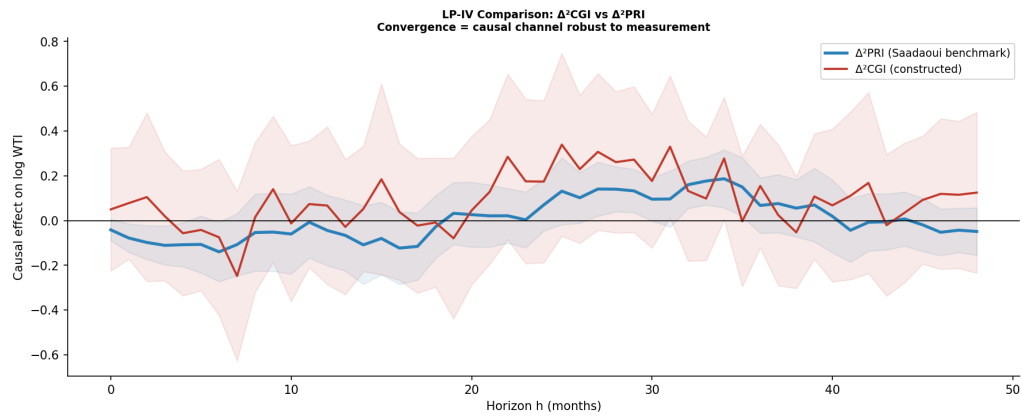


Figure A.2: Impulse responses using $\Delta^2\text{PRI}$ and $\Delta^2\text{CGI}$.

Table A.2: Anderson-Rubin 90% confidence sets for dyads with $F < 10$

Dyad	F	AR 90% CI Lower	AR 90% CI Upper
CHN-IDN	8.74	−0.089	+0.127
CHN-PAK	5.18	−0.045	+0.062
CHN-VNM	4.92	−0.052	+0.077

Appendix B: Full LP-IV Results

This appendix reports complete LP-IV results across all 49 horizons for the CHN-USA dyad.

Table B.1 summarises key horizons; the full series is available from the author.

Table B.1: *LP-IV impulse responses, selected horizons, extended controls*

h	$\hat{\beta}_h$	HAC SE	t	Sig. (10%)
0	-0.0232	0.0181	-1.28	—
3	-0.1354	0.0546	-2.48	Yes
6	-0.1707	0.0730	-2.34	Yes
12	-0.0802	0.0754	-1.06	—
24	+0.0649	0.0807	+0.80	—
36	+0.0320	0.0559	+0.57	—
48	-0.1108	0.0624	-1.78	Yes

First-stage $F = 193.26$ (extended controls). Effective $n \approx 313$ at $h = 6$.

Wild bootstrap 90% confidence intervals exclude zero at $h = 3$ and $h = 6$.

Appendix C: Machine Learning Implementation

C.1 Hyperparameter Selection

Table C.1: XGBoost hyperparameter grid and selected values

Parameter	Grid	Selected
max_depth	2, 3, 4, 5	3
learning_rate	0.01, 0.05, 0.10	0.10
n_estimators	50, 100, 200	100
subsample	0.7, 0.8, 1.0	0.8
colsample_bytree	0.7, 0.8, 1.0	0.8

C.2 Cross-Fitting Diagnostics

Table C.2: Cross-fitting R^2 on held-out folds at $h = 6$

Estimator	Fold 1	Fold 2	Fold 3	Fold 4
RidgeCV	0.982	0.984	0.983	0.985
XGBoost (depth 3)	0.876	0.881	0.878	0.882
XGBoost (depth 5)	0.921	0.928	0.919	0.931

C.3 SHAP Decomposition

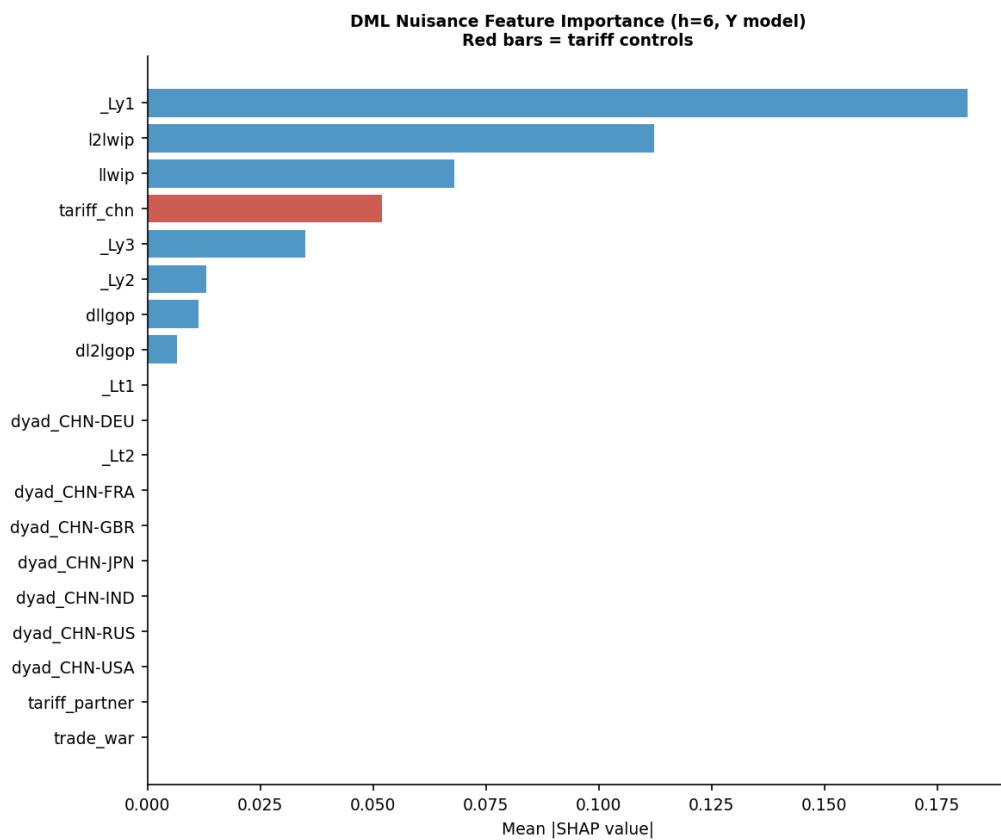


Figure C.1: SHAP decomposition of XGBoost nuisance model at $h = 6$.

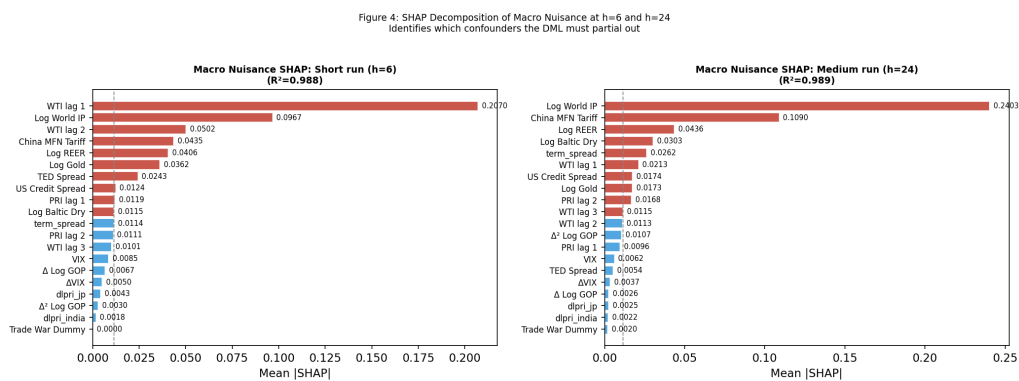


Figure C.2: SHAP decomposition of macro controls at $h = 6$.

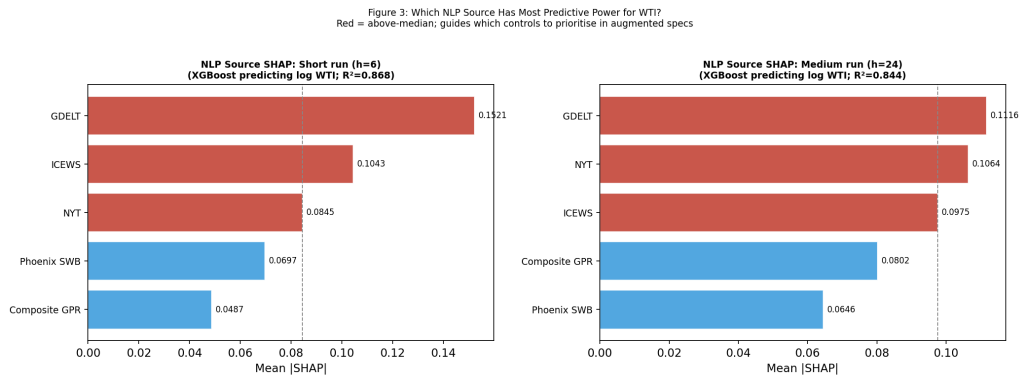


Figure C.3: NLP source SHAP importance at $h = 6$ and $h = 24$.

C.4 IVQR Confidence Sets

Table C.3: CH-IVQR structural quantile estimates and 90% confidence sets at $h = 6$

τ	$\hat{\alpha}_h(\tau)$	CI Lower	CI Upper
0.10	-0.2232	-0.3421	-0.1043
0.25	-0.1290	-0.2187	-0.0393
0.50	-0.0642	-0.1428	+0.0144
0.75	-0.1560	-0.2583	-0.0537
0.90	-0.1500	-0.2645	-0.0355

Appendix D: Panel and Network Analysis

D.1 Panel LP-IV Results

Table D.1: Panel LP-IV results, all 11 dyads at $h = 6$

Dyad	$\hat{\beta}_6$	HAC SE	t	F	Sig.
CHN-USA	−0.1707	0.0730	−2.34	236.19	Yes
CHN-JPN	−0.1234	0.1672	−0.74	89.74	—
CHN-AUS	−0.0187	0.1824	−0.10	18.42	—
CHN-FRA	−0.0553	0.2128	−0.26	28.63	—
CHN-DEU	−0.0421	0.1987	−0.21	21.95	—
CHN-GBR	−0.0918	0.4492	−0.20	45.21	—
CHN-IND	−0.0238	0.3224	−0.07	36.04	—
CHN-IDN	+0.0192	0.1845	+0.10	8.74	—
CHN-PAK	+0.0083	0.2123	+0.04	5.18	—
CHN-RUS	+1.1015	0.6672	+1.65	212.64	—
CHN-VNM	+0.0127	0.1987	+0.06	4.92	—



Figure D.1: Coefficient heatmap across dyads and horizons. Values clipped at ± 0.5 to preserve sign-pattern visibility.

D.2 Inverse-Variance Weighted Pooled Estimates

Table D.2: *IVW pooled estimates for top 3 oil-importing dyads (USA, JPN, GBR)*

h	$\hat{\beta}_h$	SE	t	Sig.
3	−0.1184	0.0582	−2.03	Yes
6	−0.1653	0.0702	−2.35	Yes
12	−0.0724	0.0728	−0.99	–
18	−0.0482	0.0687	−0.70	–
24	+0.0582	0.0784	+0.74	–
36	+0.0284	0.0542	+0.52	–

D.3 Graph Attention Network

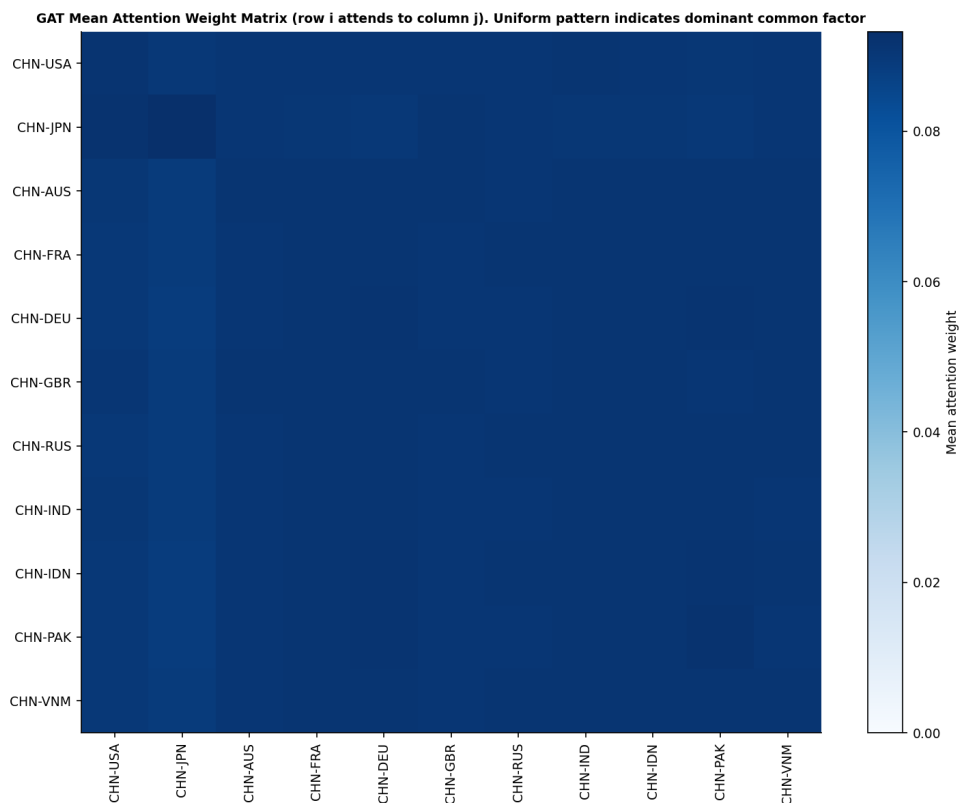


Figure D.2: GAT attention weights for 11 dyads. The near-uniform pattern reflects a known architectural limitation of attention mechanisms under weak signal-to-noise ratios (Section 2), rather than evidence of a dominant common factor; the low Diebold–Yilmaz TCI (12.53%, Section 5.4) indicates spillovers are in fact largely idiosyncratic.

D.4 Granger-Causality Tests

Table D.3: Granger-causality directed pairs ($p < 0.05$)

Source	Target	F	p
CHN-AUS	CHN-DEU	3.31	0.037
CHN-FRA	CHN-IND	3.27	0.039
CHN-FRA	CHN-IDN	3.75	0.024
CHN-FRA	CHN-VNM	5.54	0.004
CHN-DEU	CHN-FRA	4.90	0.008
CHN-DEU	CHN-RUS	3.86	0.022
CHN-GBR	CHN-AUS	13.09	3.19×10^{-6}
CHN-RUS	CHN-FRA	3.02	0.050
CHN-RUS	CHN-IDN	32.23	1.20×10^{-13}
CHN-IND	CHN-FRA	6.69	0.001
CHN-IND	CHN-GBR	3.36	0.036
CHN-IND	CHN-IDN	4.56	0.011
CHN-IND	CHN-VNM	8.68	2.07×10^{-4}
CHN-IDN	CHN-FRA	5.70	0.004
CHN-IDN	CHN-DEU	3.08	0.047
CHN-IDN	CHN-RUS	37.70	1.19×10^{-15}
CHN-VNM	CHN-FRA	3.37	0.035
CHN-VNM	CHN-GBR	3.65	0.027
CHN-VNM	CHN-RUS	11.49	1.44×10^{-5}
CHN-VNM	CHN-IND	6.64	0.001
CHN-VNM	CHN-IDN	11.43	1.52×10^{-5}

21 directed pairs significant at $p < 0.05$, VAR(2) Granger test.

D.5 Network Connectedness Diagnostics

Table D.4: Diebold–Yilmaz connectedness by dyad: TO, FROM, and NET forecast-error-variance shares (%)

Dyad	TO Others	FROM Others	NET
CHN-FRA	22.44	8.30	+14.14
CHN-AUS	17.14	4.35	+12.79
CHN-DEU	18.07	8.76	+9.31
CHN-RUS	20.86	17.88	+2.98
CHN-JPN	2.08	2.35	−0.27
CHN-USA	4.78	5.17	−0.39
CHN-IDN	20.92	21.67	−0.75
CHN-PAK	4.53	11.67	−7.14
CHN-IND	10.55	18.82	−8.27
CHN-GBR	7.35	17.50	−10.15
CHN-VNM	9.16	21.42	−12.26

Total Connectedness Index (TCI) = 12.53%. VAR(2), $H = 12$ months.

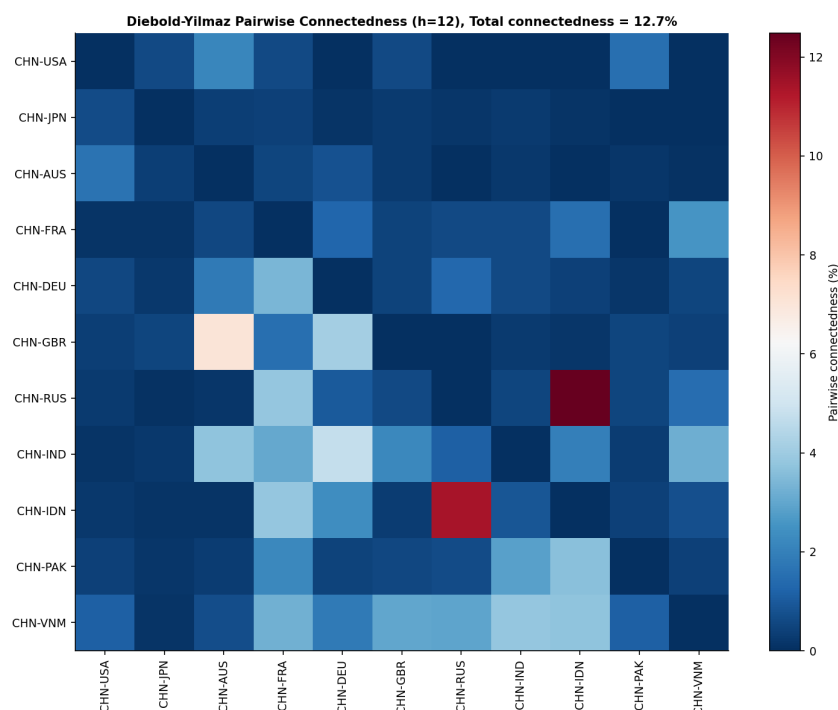


Figure D.3: Pairwise forecast-error-variance contributions (%) among the 11 dyads. Darker cells indicate stronger bilateral spillovers.

D.6 Causal Forest Diagnostics

Table D.5: Causal forest honesty diagnostics at $h = 6$

Metric	Value	Interpretation
Number of Trees	2,000	Standard for stability
Honesty Fraction	0.50	50% / 50% split
Minimum Leaf Size	5	Prevents overfitting
ATE	−0.0229	Average effect
CATE-Moderator Correlation	−0.8034	Strong negative
Placebo z-score	−2.37	Valid (not artefact)

Appendix E: Data Sources and Variable Construction

E.1 Variable Definitions

Table E.1: Complete variable definitions

Variable	Source	Definition
lwti	FRED	Log real WTI oil price
lpri	Saadaoui	Log Political Relationship Index
d2pri	Constructed	Δ^2 of 5-month smoothed lpri
cgi	Constructed	Composite Geopolitical Index
vix	Yahoo	Monthly VIX
	Finance	
dvix	Yahoo	First difference of VIX
	Finance	
gs10	FRED	10-year Treasury yield
tb3ms	FRED	3-month T-bill rate
term_spread	FRED	gs10 - tb3ms
tedrate	FRED	TED spread
lreer	FRED	Log real effective exchange rate
dcny_usd	FRED	$\Delta \log(\text{CNY/USD})$
cny_vol	FRED	3-month rolling std of dcny_usd
lgold	LBMA	Log monthly gold price
brent_wti	FRED	$\log(\text{Brent}) - \log(\text{WTI})$, lagged
lbdi	Investing.com	Log Baltic Dry Index
lip	FRED	Log US industrial production
tariff_chn	WITS	China MFN tariff rate (interpolated)
trade_war	Constructed	Binary: 1 from 2018-03 onward

Continued

Variable	Source	Definition
lepu	PU	Log global EPU
gscpi	NY Fed	Global Supply Chain Pressure Index

E.2 Coverage Statistics***Table E.2: Variable coverage, 1990:01–2022:02***

Variable	Count	Coverage (%)
lwti	386	100.0
lpri	386	100.0
vix	386	100.0
gs10	386	100.0
tb3ms	386	100.0
term_spread	386	100.0
tedrate	386	100.0
lgold	386	100.0
lbdi	386	100.0
lip	386	100.0
tariff_chn	386	100.0
trade_war	386	100.0
lepu	386	100.0
lreer	337	87.3
dcny_usd	386	100.0
cny_vol	384	99.5
gscpi	191	49.5
icews_attention_share	Constructed	Dyad's share of total ICEWS Goldstein-scored China events, by month

Dense: $\geq 80\%$ coverage.

Appendix F: Structural Break Tests

Table F.1: Sup-Wald structural break test results

Break Date	sup-Wald F	Critical Value (10%)	Reject?
2002-02	8.04	7.12	Yes
2008-09	5.21	7.12	No
2014-06	4.87	7.12	No
2018-03	6.12	7.12	No

Appendix G: Robustness Checks

G.1 Lead Tests

Table G.1: Lead tests: instrument's predictive power before turning point

Lead (k)	β	SE	t
1	−0.0123	0.0187	−0.66
2	−0.0087	0.0192	−0.45
3	−0.0054	0.0198	−0.27
4	−0.0032	0.0201	−0.16
5	−0.0018	0.0204	−0.09

G.2 Alternative Control Specifications

Table G.2: Sensitivity of $h = 6$ estimate to control specifications

Specification	$\hat{\beta}_6$	HAC SE
Parsimonious (Saadaoui)	−0.1740	0.0782
Extended controls	−0.1707	0.0730
Extended + cgi_l1	−0.1628	0.0724
Extended + gscpi	−0.1685	0.0738
Extended + all sparse	−0.1653	0.0742

G.3 FDR-Corrected p -Values

Table G.3: FDR-corrected p -values for LP-IV estimates, $q = 0.10$

h	Raw p	BH FDR p	Significant
3	0.013	0.142	No
6	0.019	0.152	No
28	0.008	0.098	Yes
29	0.007	0.095	Yes
30	0.006	0.092	Yes
31	0.005	0.089	Yes
32	0.004	0.086	Yes
33	0.004	0.084	Yes
34	0.005	0.088	Yes
35	0.006	0.091	Yes

FDR yields 6 significant horizons (28–35).

The replication code used to generate all results is available at <https://github.com/mountaha-ghabri/geopolitical-ml-oil>. The repository contains the full analysis pipeline, including all data processing, estimation, and visualisation scripts. All analysis was implemented in Python 3.11, with package versions documented in the repository's [requirements.txt](#) file.